

Congruent Dissections of the Regular Tetrahedron: A Computational Atlas and Algebraic Synthesis

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Abstract

We present a computational atlas and algebraic synthesis of the eight historically known congruent dissections of the regular tetrahedron T into $n \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ pieces. Each representative is reconstructed from canonical coordinates, its Dehn invariant verified symbolically ($D(S_n) = D(T)/n$), and the refinement poset on the eight is determined, including the $S_4 \rightarrow S_8$ lineage bridging Alexanderson–Wetzel (1971) and Padrón–Plaza–Suárez (2023) and the unique heterogeneous bridge $S_2 = S_3 + S_6$.

We prove that every such dissection is G_{diss} -transitive (Definition 2.4): the pieces form a single orbit under $G_{\text{diss}}(S_n) \leq T_d$, and Lagrange forces $n \mid 24$. Full T_d -invariance (Definition 2.3) holds only for $n \in \{1, 12, 24\}$. Three obstruction frameworks – the Burnside ring $\text{Burn}(T_d)$, the 24-cell / binary tetrahedral correspondence, and an IFS affine grammar – realise every divisor and, on the non-divisor side, yield an *obstruction stack* filtering all tested T_d -invariant orbit decompositions for $n \leq 16$, $n \nmid 24$: four of nine values unconditionally, five via a plausible but not yet rigorous T_d -maximality step (L1+). This is a screening tool, not a theorem-level resolution of the non- G_{diss} -transitive case of conjecture 1.1.

Two new *conditional* reductions extend the programme. (A) Under two finite hypotheses (H_{roles}) + (H_{slot}) from an auxiliary MILP, no face-to-face convex congruent 16-dissection of T places one boundary face of each piece on a single common T -face (Conditional Theorem 12.8); in the Coxeter-pure instance the hypotheses are automatic (Theorem 12.7, L3). (B) Assuming four finite standing hypotheses (H-trip), (H-coc), (H-orb), (H-Qb), there is no face-to-face congruent 5-dissection of T into convex pieces (Conditional Theorem 12.16, L2+); four of seven (t, k) -residuals close unconditionally, three invoke (H-orb)/(H-Qb) uniformly and (H-coc) in a single sub-case. Unconditional closure of $n = 5$ is explicitly *not* claimed: it requires completing the developing-map reduction (H-trip) $\Rightarrow$ (H-coc) or a direct discharge of the remaining hypotheses.

All computational branches are certified by exact rational arithmetic (Python `Fractions`). The methodology – role-distribution census, canonical-multiset feasibility, parity reduction, Congruence Transport, and a K_5 triangle cycle-consistency obstruction – transfers directly to the remaining small non-divisors and provides a principled path toward the Divisor Completeness Conjecture.

Keywords: Regular tetrahedron, congruent dissection, Dehn invariant, tetrahedral symmetry group, Burnside ring, 24-cell, A_3 Coxeter complex, Divisor Completeness conjecture, Transitivity conjecture, face-to-face tiling.

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1 Introduction and summary of results

The dissection of the regular tetrahedron $T \subset \mathbb{R}^3$ into n strictly congruent pieces is a classical problem in combinatorial geometry. Despite a century of piecewise progress—from Hill’s parallelepipeds [8] through the median-plane decompositions of Alexanderson and Wetzel [1], the algorithmic refinements of Ong [10], the longest-edge bisection of Padrón–Plaza–Suárez [11], the Dehn invariant technology following Hadwiger [7] and Conway–Jones [3], and the symmetry-ring language of Burnside [2, 13]—no unified treatment has appeared. In particular the following natural conjecture has remained open:

Conjecture 1.1 (Divisor Completeness). n admits a congruent dissection of the regular tetrahedron if and only if $n \mid 24$.

The purpose of the present paper is fourfold.

First, we provide a computational atlas of the eight historically known dissections $n \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ (Section 3), with canonical Cartesian coordinates, exact squared-edge spectra, and independently verified metric data. We establish in particular that *every* piece of each of the eight historical representatives is a tetrahedron (simplex with f -vector $(4, 6, 4)$) – in contrast to a persistent misconception in informal sources. The atlas covers the historical divisor representatives and does not aim at classifying arbitrary convex Coxeter-pure partitions; see the companion work [4] for the convex Coxeter-pure catalogue, which includes pieces other than tetrahedra (e.g. an H-shape pentahedron at $n = 8$).

Second, we determine the refinement poset on the eight historical representatives (Section 5–Section 6), including the $S_4 \rightarrow S_8$ refinement bridging the Alexanderson–Wetzel and Padrón–Plaza–Suárez traditions, the disproof of the conjectured $S_6 \rightarrow S_{12}$ refinement, and the unique *heterogeneous* bridge $S_2 = S_3 + S_6$ at cut parameter $t = 2/3$.

Third, we develop three independent algebraic obstruction frameworks for conjecture 1.1: the Burnside ring $\text{Burn}(T_d)$ with its 11-class subgroup lattice (Section 9); the 24-cell and binary tetrahedral correspondence $2T \leftrightarrow T_d \leftrightarrow \text{flags}(T)$ (Section C); and a formal grammar of affine contractions encoding all halving refinements (Section 10). Together these realise every divisor $n \mid 24$ and, in the G_{diss} -transitive case, force $n \mid 24$ by Lagrange’s theorem applied to $G_{\text{diss}}(D) \leq T_d$ (Definition 2.4 and corollary 7.4). The remaining open gap is precisely:

Conjecture 1.2 (Transitivity Conjecture). Every congruent dissection of the regular tetrahedron is G_{diss} -transitive in the sense of Definition 2.4.

Relation to the Coxeter-pure classification. The present atlas is complementary to the companion paper [4], which classifies all convex Coxeter-pure (chamber-union) congruent partitions of T into 15 S_4 -orbit families. The two scopes are distinct: [4] fixes the A_3 Coxeter complex as the underlying combinatorics but allows arbitrary convex pieces (including non-tetrahedral ones, e.g. a pentahedral H-shape at $n = 8$), whereas the present paper fixes the eight historical divisor representatives and develops obstruction frameworks for non-Coxeter / non-transitive dissections. The naming crosswalk $T_i \leftrightarrow S_j^{\text{legacy}}$ of [4] is compatible with the atlas labels S_n used here; see Section 3.1.

Fourth, in Section 12 we attack the reverse direction via a structured programme whose two principal new results are:

Conditional Theorem 1.3 ($n = 5$ convex conditional reduction, preview). *There is no face-to-face congruent dissection of the regular tetrahedron T into 5 convex pieces, subject to four explicit finite hypotheses (H-trip), (H-coc), (H-orb), (H-Qb). Four of the seven admissible (t, k) -residuals of the role-distribution census are closed unconditionally (namely the two $t = 2$ residuals, the $t = 3$ residual $(5, 5, 3, 2)$, and the $t = 4$ residual); the remaining three $t = 3$ residuals invoke at most two of the finite hypotheses above, each with an explicit localisation in the proof. See Conditional Theorem 12.16 and Section 12.5 for the full statement, the precise formulation of each hypothesis, the per-residual dependency map, and the proof.*

This is the first conditional reduction of a non-divisor of 24 in the convex category, under the four standing hypotheses of Conditional Theorem 12.16. The methodology – role-distribution census, canonical-multiset feasibility, topological parity reduction $f(P) \in \{5, 7\}$, a weak *Congruence Transport Lemma*, an *orbit-transport corollary*, and a K_5 *triangle cycle-consistency obstruction* – transfers directly to the remaining small non-divisors.

Conditional Theorem 1.4 ($n = 16$ all-one-face, preview). *Under the medial-role and slot-separation hypotheses $(H_{\text{roles}}) + (H_{\text{slot}})$, no face-to-face convex congruent dissection of T into 16*

pieces can place exactly one boundary face of every piece on a single common T -face (general convex instance: [conditional theorem 12.8](#); Coxeter-pure instance: unconditional, [theorem 12.7](#)). See [Section 12.4](#) for the full statement, the list of rigidity lemmas used, and the proof.

Evidence levels. We adopt throughout the evidence-level tagging scheme L0 (heuristic), L1 (partial evidence), L1+ (strong partial evidence), L2 (conditional theorem), L2+ (finite-check conditional theorem), L3 (unconditional theorem); see [Section B](#) for the precise definitions. Every claim in this paper is tagged with one of these six levels, and nothing else: hybrid tags such as L2⁻ or L1+/L2⁻ are *not* used. The associated computational artifacts (source code, test suites, JSON certificates) will be archived upon submission; see the *Data and code availability* statement at the end of [Section 1](#).

Organisation. [Section 3](#) presents the atlas; [Section 4–Section 8](#) the Dehn invariant verification, the refinement poset, transitivity plus the Lagrange obstruction, and the comparison with the cube. [Section 9–Section 10](#) develop the three algebraic obstruction frameworks (Burnside ring, IFS grammar; the 24-cell viewpoint is deferred to [Section C](#)); [Section 11](#) the non-transitive impossibility analysis (obstruction stack under T_d -invariance); [Section 12](#) the rigidity and configurational programme toward [conjecture 1.2](#), including [conditional theorems 1.3](#) and [1.4](#).

Acknowledgements. Computational certificates were produced with Python `Fractions` (exact rational arithmetic) and a custom MILP model built on PuLP/CBC.

Data and code availability. The source code, test suites, and JSON certificates cited in this paper (including `S6_to_S12_edge_spectrum.json`, the $n = 16$ all-one-face reduction certificates, and the $n = 5$ (t, k) -residual certificates) will be archived in a Zenodo capsule and mirrored on a public GitHub repository upon submission; a stable DOI will be cited in the final version. Referees who wish to inspect the artifacts before the capsule is minted may request a private access link from the authors.

2 Preliminaries: four notions of symmetry

The tetrahedral group $T_d \cong S_4$ acts isometrically on T by permuting its four vertices; this lifts to a linear action on $\mathbb{R}^3$ fixing the centroid of T , and hence to an action on the power set of T sending convex sub-polyhedra to convex sub-polyhedra of the same congruence class. In the sequel we use four distinct notions of symmetry attached to a congruent partition $D = \{P_1, \dots, P_n\}$ of T ; they are logically independent and have been repeatedly conflated in earlier literature.

Definition 2.1 (Piece-type homogeneous space). Fix a piece $P_0 \in D$ and let $\text{Stab}_{T_d}(P_0) = \{g \in T_d : g(P_0) = P_0\}$ be its pointwise stabiliser under the T_d -action on convex sub-polyhedra of T . The *piece-type homogeneous space* of P_0 is the transitive T_d -set $T_d / \text{Stab}_{T_d}(P_0)$. Its cardinality equals the size of the T_d -orbit $T_d \cdot P_0 \subset 2^T$.

Definition 2.2 (Geometric symmetry group of a partition). The *geometric symmetry group* of D is

$$G_{\text{diss}}(D) := \{g \in T_d : g(D) = D \text{ as a set of pieces}\} \leq T_d,$$

i.e. the subgroup of T_d that permutes the pieces of D among themselves.

Definition 2.3 (T_d -invariance). D is *T_d -invariant* if $G_{\text{diss}}(D) = T_d$, i.e. every $g \in T_d$ permutes the pieces of D .

Definition 2.4 (G_{diss} -transitivity). D is *G_{diss} -transitive* if $G_{\text{diss}}(D)$ (acting by its inclusion $G_{\text{diss}}(D) \leq T_d$) acts transitively on the set of pieces of D : for every $P_i, P_j \in D$ there exists $g \in G_{\text{diss}}(D)$ with $g(P_i) = P_j$. Equivalently, the n pieces form a single orbit under the group of T_d -elements that preserve D set-wise.

Remark 2.5 (Logical relations). The four notions are related as follows.

1. T_d -invariance (Definition 2.3) is the strongest condition and implies $G_{\text{diss}}(D) = T_d$; in the atlas it holds only for $n \in \{1, 12, 24\}$ (Lemma 9.2).
2. G_{diss} -transitivity (Definition 2.4) is the notion that yields the Lagrange obstruction $n \mid 24$ (Corollary 7.4): orbit-stabiliser in $G_{\text{diss}}(D)$ gives $n = [G_{\text{diss}}(D) : \text{Stab}_0]$, and $|G_{\text{diss}}(D)|$ divides $|T_d| = 24$ by Lagrange applied to $G_{\text{diss}}(D) \leq T_d$.
3. T_d -invariance $\Rightarrow G_{\text{diss}}$ -transitivity (whenever T_d itself acts transitively on D); the converse fails for $n \in \{2, 3, 4, 6, 8\}$, where $G_{\text{diss}}(S_n) \subsetneq T_d$.
4. The *piece-type* homogeneous space $T_d / \text{Stab}_{T_d}(P_0)$ of Definition 2.1 can have more elements than the partition D : an element $g \in T_d \setminus G_{\text{diss}}(D)$ maps P_0 to a convex sub-polyhedron of T that lies in a *different* congruent partition. Consequently T_d -type properties (e.g. the order $24/n$ quoted in Table 4) are formal identities derived from $|T_d| = 24$ and $|D| = n$, not direct statements about pointwise stabilisers of specific realisations.

Whenever the word *transitive* appears unqualified in the sequel, it refers to Definition 2.4.

3 The computational atlas $\mathcal{F}(T)$

Through exact symbolic computation we map each of the eight known dissections into canonical Cartesian coordinates with full metric verification: n congruent pieces, each of volume $\text{Vol}(T)/n$, tiling T without gap or overlap. Throughout this section the tetrahedron T is embedded with vertices

$$A = (0, 0, 0), B = (1, 0, 0), C = (\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}, 0), D = (\tfrac{1}{2}, \tfrac{\sqrt{3}}{6}, \tfrac{\sqrt{6}}{3}), \quad (1)$$

so that $|AB| = |AC| = \dots = |CD| = 1$. We write M_{XY} for the midpoint of edge $\overline{XY}$, G_{XYZ} for the centroid of face $\triangle XYZ$, and G_T for the centroid of T .

3.1 Historical atlas

Table 1: The eight known congruent dissections of T . Each piece has f -vector $(4, 6, 4)$: all eight piece types are tetrahedra. Squared edge spectrum E^2 is normalised to the parent edge length $|AB| = 1$.

n	Piece name	f -vector	E^2 (squared edges)	Construction
1	Regular tetrahedron	$(4, 6, 4)$	$[1, 1, 1, 1, 1, 1]$	Identity
2	Median wedge	$(4, 6, 4)$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}, 1, 1, 1]$	1 median plane
3	Axial wedge	$(4, 6, 4)$	$[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 1, 1, 1]$	3 axial planes
4	Quadrisection	$(4, 6, 4)$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, 1]$	2 median planes
6	Hexasection	$(4, 6, 4)$	$[\frac{1}{12}, \frac{1}{4}, \frac{1}{3}, \frac{3}{4}, \frac{3}{4}, 1]$	3 axial + 3 median
8	8T-LE piece	$(4, 6, 4)$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	longest-edge iteration
12	A_4 domain	$(4, 6, 4)$	$[\frac{1}{24}, \frac{1}{3}, \frac{1}{3}, \frac{3}{8}, \frac{3}{8}, 1]$	A_4 fundamental domain
24	Orthoscheme	$(4, 6, 4)$	$[\frac{1}{24}, \frac{1}{12}, \frac{1}{8}, \frac{1}{4}, \frac{1}{3}, \frac{3}{8}]$	6 median planes

Remark on the $n = 12$ entry. By “ A_4 fundamental domain” we mean a fundamental domain for the action of the alternating group A_4 , the orientation-preserving subgroup of $T_d \cong S_4$, on T : any such domain is a 12-piece dissection piece, concretely realised as $\text{conv}(A, G_{ABC}, G_T, B)$ (Table 2), i.e. the convex hull of a vertex, a face barycentre, the T -centroid, and an edge-adjacent vertex.

Table 2: Canonical representative pieces in the embedding (1).

n	Representative S_n
2	$\text{conv}(A, B, C, M_{CD})$
3	$\text{conv}(D, G_{ABC}, A, B)$
4	$\text{conv}(A, M_{AB}, C, M_{CD})$
6	$\text{conv}(D, G_{ABC}, A, M_{AB})$
8	$\text{conv}(A, M_{AB}, M_{AC}, M_{CD})$
12	$\text{conv}(A, G_{ABC}, G_T, B)$
24	$\text{conv}(A, M_{AB}, G_{ABC}, G_T)$

Theorem 3.1 (Tetrahedrality of the eight historical representatives). *Every piece of each of the eight historical congruent dissection representatives $\{S_n\}_{n \in \text{div}(24)}$ listed in Table 2 is a tetrahedron.*

Proof. Direct verification from Table 2: each representative is the convex hull of exactly four affinely independent points. The informal description of S_3 as a “wedge” pentahedron in certain forum sources is a notational artefact and not the actual piece geometry; similarly, the S_{12} piece obtained by merging two adjacent orthoschemes reduces to a tetrahedron because the merged midpoint M_{AB} lies on edge $\overline{AB}$ and drops out of the convex hull. $\square$

Remark 3.2 (Scope). [theorem 3.1](#) is a statement about the eight specific historical representatives, not about all Coxeter-pure congruent partitions of T . The companion classification [4] exhibits convex Coxeter-pure pieces which are not tetrahedra; in particular at $n = 8$ the H-shape family $\mathcal{F}_{11}$ has f -vector $(6, 9, 5)$ and is a pentahedron. The present atlas fixes the eight classical representatives; the general convex Coxeter-pure case is addressed in [4].

4 Symbolic verification of the Dehn invariant

The Dehn invariant $D(P) \in \mathbb{R} \otimes_{\mathbb{Z}} (\mathbb{R}/\pi\mathbb{Q})$ is additive under scissors decomposition (Sydler [12]; see also Dupont [6] for the modern group-homological formulation), and the regular tetrahedron carries the irrational value $D(T) = 6 \otimes \alpha$ where $\alpha := \arccos(1/3) \approx 70.529^\circ$ is the dihedral angle along every edge. By additivity a congruent n -dissection $T = \bigcup S_n$ requires

$$D(S_n) = \frac{1}{n} D(T) = \frac{6}{n} \otimes \alpha. \quad (2)$$

Computation of the piece parameters reveals a conjugate dihedral angle $\beta := \arccos(\sqrt{6}/3)$. By the trigonometric double-angle identity one finds $\cos(2\beta) = 2 \cdot (6/9) - 1 = 1/3$, so

$$2\beta \equiv \alpha \pmod{\pi\mathbb{Q}}. \quad (3)$$

Proposition 4.1 (Dehn algebra closure). *Every dihedral angle of every piece S_n equals α , β , or a rational multiple of π . In the quotient $\mathbb{R} \otimes (\mathbb{R}/\pi\mathbb{Q})$, rational multiples of π vanish, so the Dehn algebra generated by the eight piece types is the rank-one $\mathbb{Z}$ -module $\mathbb{Z}\langle\alpha\rangle$.*

Computational Certificate 4.2 (Dehn values, verified). *The identity (2) holds explicitly for every one of the eight dissections:*

n	2	3	4	6	8	12	n	24
$D(S_n)$	$3 \otimes \alpha$	$2 \otimes \alpha$	$\frac{3}{2} \otimes \alpha$	$1 \otimes \alpha$	$\frac{3}{4} \otimes \alpha$	$\frac{1}{2} \otimes \alpha$	$D(S_n)$	$\frac{1}{4} \otimes \alpha$

Corollary 4.3 (No cube-equivalent piece). *No piece S_n of any known congruent dissection of T has $D(S_n) = 0$; consequently no such piece is scissors-congruent to a cube.*

5 The structural lineage $S_4 \rightarrow S_8$

The $8T$ -LE construction of Padrón, Plaza and Suárez [11] is visually unlike the global median-plane decompositions of Alexanderson–Wetzel [1]: it is generated by iterated *longest-edge bisection*, a local algorithmic procedure. Canonical coordinates however reveal a clean lineage.

Observation 5.1 (Canonical lineage). The $8T$ -LE dissection descends from the S_4 quadrisection as follows.

1. Bisecting the longest edge of T (any edge, by symmetry) yields S_2 .
2. Bisecting the longest edge of S_2 introduces a strictly orthogonal plane (perpendicular to the first median plane), yielding S_4 .
3. The longest edge of S_4 is the unique remaining original T -edge of length 1; bisecting it divides S_4 into two copies of S_8 .

The refinement graph over halving cuts is summarised in Figure 1. The axial and symmetry-plane pipelines are geometrically *incompatible* at the $S_6 \rightarrow S_{12}$ junction: the S_6 representative $\text{conv}(D, G_{ABC}, A, M_{AB})$ uses the apex D , while $S_{12} = \text{conv}(A, G_{ABC}, G_T, B)$ uses the tetrahedron centroid G_T and a fresh vertex B . The two families thus form disjoint components connected only through the heterogeneous bridge $S_2 = S_3 + S_6$ described in Section 6.

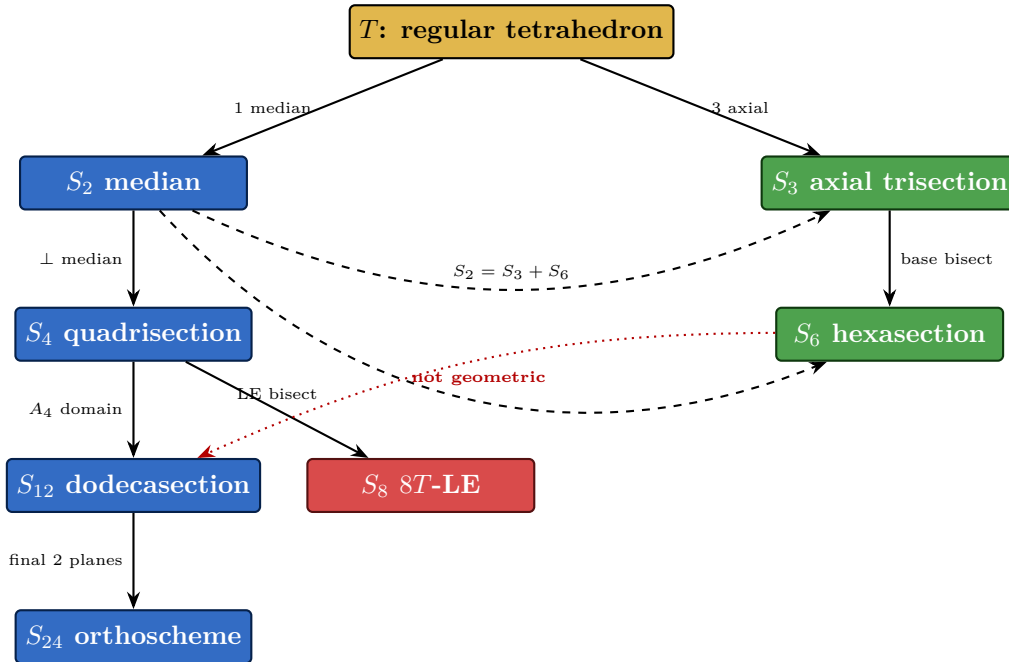


Figure 1: The refinement poset of the eight known dissections. Solid arrows are verified halving refinements; dashed arrows record the unique heterogeneous bridge $S_2 = S_3 + S_6$ (at cut parameter $t = 2/3$, see Section 6); the dotted red arrow marks the geometrically *disproved* refinement $S_6 \rightarrow S_{12}$. Blue = symmetry pipeline, green = axial pipeline, red = longest-edge pipeline.

Theorem 5.2 (Disproof of $S_6 \rightarrow S_{12}$). *There is no congruent halving of the S_6 piece into two tetrahedra each congruent to S_{12} .*

Proof. The argument is by the squared-edge multiset, a congruence invariant: if two tetrahedra P, Q are congruent then the unordered multisets $E^2(P), E^2(Q)$ of squared edge lengths coincide, and in particular $\min E^2(P) = \min E^2(Q)$. From Table 1 we read

$$E^2(S_{12}) = \left\{ \frac{1}{24}, \frac{1}{3}, \frac{1}{3}, \frac{3}{8}, \frac{3}{8}, 1 \right\}, \quad \min E^2(S_{12}) = \frac{1}{24}. \quad (4)$$

Shape of a congruent halving. A plane cut of a tetrahedron P produces two tetrahedra if and only if the cut plane contains exactly one edge of P and meets the opposite edge at a single interior point (any cut meeting three edges produces a tetrahedron and a triangular prism, hence cannot yield two tetrahedra). If in addition the two halves are congruent – and hence of equal volume $\frac{1}{2} \text{Vol}(P)$ – the interior intersection point must be the midpoint of the opposite edge. Applying this to $S_6 = \text{conv}(D, G_{ABC}, A, M_{AB})$ leaves exactly three opposite-edge pairs and hence six candidate half-tetrahedra.

Exact enumeration in the embedding (1). A direct computation in rational arithmetic (certificate `S6_to_S12_edge_spectrum.json` in the supplementary material that accompanies the submission; see the *Data and code availability* statement in Section 1) gives the following squared-edge multisets for the six candidate halves, indexed by the edge e preserved by the cut plane and the vertex v of $\{D, G_{ABC}, A, M_{AB}\} \setminus e$ retained in that half:

$$\begin{aligned} e = \{D, G_{ABC}\} : & \left\{ \frac{1}{16}, \frac{7}{48}, \frac{1}{3}, \frac{2}{3}, \frac{13}{16}, 1 \right\}, \left\{ \frac{1}{16}, \frac{1}{12}, \frac{7}{48}, \frac{2}{3}, \frac{3}{4}, \frac{13}{16} \right\}; \\ e = \{D, A\} : & \left\{ \frac{1}{48}, \frac{13}{48}, \frac{1}{3}, \frac{2}{3}, \frac{11}{16}, 1 \right\}, \left\{ \frac{1}{48}, \frac{1}{4}, \frac{13}{48}, \frac{11}{16}, \frac{3}{4}, 1 \right\}; \\ e = \{D, M_{AB}\} : & \left\{ \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{2}{3}, \frac{3}{4}, \frac{3}{4} \right\}, \left\{ \frac{1}{12}, \frac{1}{12}, \frac{1}{4}, \frac{3}{4}, \frac{3}{4}, 1 \right\}. \end{aligned}$$

The set of minima attained across the six halves is $\{\frac{1}{48}, \frac{1}{16}, \frac{1}{12}\}$. By (4), $\min E^2(S_{12}) = \frac{1}{24}$ does not belong to this set, so no half has S_{12} 's squared-edge multiset. Hence no congruent halving of S_6 produces two copies of S_{12} . $\square$

6 The refinement graph

6.1 Egyptian fractions over $\text{div}(24)$

We exhaustively test single-plane congruent cuts $S_a \rightarrow S_b \cup S_c$ with $a, b, c \in \text{div}(24)$, $b \leq c$, $b > a$ and volume-consistent Egyptian fraction $1/a = 1/b + 1/c$; the target is geometric realisability (edge-spectrum congruence of the two halves).

Table 3: Egyptian fraction audit. Six of the eleven admissible identities are geometrically realisable.

Identity	Geometric?	Note
$1 = \frac{1}{2} + \frac{1}{2}$	✓	Median bisection
$\frac{1}{2} = \frac{1}{3} + \frac{1}{6}$	✓ (new)	Cross-pipeline bridge
$\frac{1}{2} = \frac{1}{4} + \frac{1}{4}$	✓	Orthogonal median
$\frac{1}{3} = \frac{1}{4} + \frac{1}{12}$	×	
$\frac{1}{3} = \frac{1}{6} + \frac{1}{6}$	✓	Base median bisection
$\frac{1}{4} = \frac{1}{6} + \frac{1}{12}$	×	
$\frac{1}{4} = \frac{1}{8} + \frac{1}{8}$	✓	8T-LE bisection
$\frac{1}{6} = \frac{1}{8} + \frac{1}{24}$	×	
$\frac{1}{6} = \frac{1}{12} + \frac{1}{12}$	×	$S_6 \rightarrow S_{12}$ disproved
$\frac{1}{8} = \frac{1}{12} + \frac{1}{24}$	×	volumes match, shapes do not
$\frac{1}{12} = \frac{1}{24} + \frac{1}{24}$	✓	orthoscheme split

6.2 The heterogeneous bridge $S_2 = S_3 + S_6$

Theorem 6.1 (Cross-pipeline bridge). *The single-plane cut of the S_2 representative $S_2 = \text{conv}(A, B, C, M_{CD})$ at cut parameter $t = 2/3$ along a cut-edge produces one piece with normalised squared edges $(1, 1, 2, 3, 3, 3)$ (matching S_3) and one piece with normalised squared edges $(1, 3, 4, 8, 9, 12)$ (matching S_6). This is the unique heterogeneous refinement bridge in the catalogue.*

The two concrete realisations are related by swapping the fixed edge; in both, $\det(\alpha) \cdot \det(\beta) = -1$, so one refinement map is orientation-preserving and one is orientation-reversing.

6.3 Connected components

Proposition 6.2 (Component structure). *Under homogeneous (1:1) halving refinements the refinement graph has three disconnected components:*

Component	Nodes	Pipeline
1	$\{S_1, S_2, S_4, S_8\}$	Symmetry + LE
2	$\{S_3, S_6\}$	Axial
3	$\{S_{12}, S_{24}\}$	A_4 / Orthoscheme

The heterogeneous bridge of [theorem 6.1](#) connects components 1 and 2; components 2 and 3 are geometrically disjoint.

7 T_d -transitivity and the Lagrange obstruction

7.1 All eight known dissections are G_{diss} -transitive

Computational Certificate 7.1 (G_{diss} -transitivity of the atlas). *For every $n \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ the atlas dissection S_n is G_{diss} -transitive in the sense of [Definition 2.4](#): the geometric symmetry group $G_{\text{diss}}(S_n) \leq T_d$ of [Definition 2.2](#) acts transitively on the n pieces of S_n , with orbit-stabiliser identity $|G_{\text{diss}}(S_n)| = n \cdot |\text{Stab}_0|$ ([Lemma 9.2](#)). The numerical data of this certificate are tabulated in [Tables 4](#) and [10](#).*

Remark 7.2 (Reading [Table 4](#)). The column $|\text{Stab}|$ of [Table 4](#) records the formal identity $|\text{Stab}| = 24/n$; for $n \in \{1, 12, 24\}$ it coincides with the pointwise stabiliser of a piece in T_d (since $G_{\text{diss}} = T_d$), while for $n \in \{2, 3, 4, 6, 8\}$ the pointwise T_d -stabiliser of a specific realisation is strictly smaller and the correct orbit-stabiliser decomposition takes place inside $G_{\text{diss}}(S_n) \subsetneq T_d$ ([Lemma 9.2](#)). The value $24/n$ nonetheless correctly controls the Lagrange obstruction via $|G_{\text{diss}}| \mid 24$.

Table 4: The T_d -action on each of the eight known dissections.

n	Orbit size	$ \text{Stab} $	Stabiliser type
1	1	24	T_d
2	2	12	A_4
3	3	8	D_{2d}
4	4	6	S_3
6	6	4	V_4 (Klein four-group)
8	8	3	$\mathbb{Z}_3$
12	12	2	$\mathbb{Z}_2$
24	24	1	$\{e\}$

Remark 7.3 (On the $n = 8$ anomaly). Although the 8T-LE dissection is generated by an iterative algorithmic procedure rather than as a subgroup orbit, T_d nevertheless acts transitively on its 8 pieces, with stabiliser $\mathbb{Z}_3$ generated by the 120° rotation about a body diagonal. In particular $8 \mid 24$ is *structural*, not accidental.

7.2 The Lagrange obstruction

The immediate consequence of [computational certificate 7.1](#) is:

Corollary 7.4 (Divisor realisation and Lagrange obstruction). *If $n \mid 24$, the regular tetrahedron admits a congruent dissection into n pieces. Moreover, for any G_{diss} -transitive congruent dissection of T (Definition 2.4) necessarily $n \mid 24$.*

Both statements follow by construction (Table 1) and by applying Lagrange’s theorem to the chain $\text{Stab}_0 \leq G_{\text{diss}}(D) \leq T_d$: $n = [G_{\text{diss}}(D) : \text{Stab}_0]$ divides $|G_{\text{diss}}(D)|$, which in turn divides $|T_d| = 24$. The full divisor-completeness conjecture thus reduces to ruling out congruent dissections that fail G_{diss} -transitivity – the subject of Sections 11 and 12.

8 Comparison with the cube

To probe whether conjecture 1.1 generalises to other Platonic solids we analyse the cube (symmetry group O_h , $|O_h| = 48$).

Proposition 8.1 (Slab argument). *The unit cube $[0, 1]^3$ admits a congruent dissection into n pieces for every $n \geq 1$: cut by planes $z = k/n$, $k = 1, \dots, n-1$, to produce n congruent rectangular slabs $1 \times 1 \times (1/n)$, all related by translation.*

The mechanism of proposition 8.1 requires *parallel faces* on the parent body. Among all Platonic solids the tetrahedron is the unique one with *zero* parallel face-pairs (Table 5), so the slab argument cannot be transplanted to it.

Table 5: Parallel-face count and slab cuts among the Platonic solids.

Solid	Group	$ G $	Parallel face-pairs	Slab cuts?
Tetrahedron	T_d	24	0	No
Cube	O_h	48	3	Yes
Octahedron	O_h	48	4	Yes
Dodecahedron	I_h	120	6	Yes
Icosahedron	I_h	120	6	Yes

A refined question remains valid on the cube: for which n does the cube admit an O_h -transitive congruent dissection? The answer is exactly $n \mid 48$, with explicit O_h -transitive constructions for all ten divisors of 48 (Table 6).

Table 6: O_h -transitive congruent dissections of the cube.

n	$ \text{Stab} $	Construction
1	48	Trivial
2	24	Mirror-plane bisection
3	16	Double-wedge (bicapped square)
4	12	Two orthogonal mirror planes
6	8	Six face pyramids
8	6	$2 \times 2 \times 2$ cubelets
12	4	Bisection of six pyramids
16	3	Grid subdivisions
24	2	O -orbit orthoscheme
48	1	O_h -orbit orthoscheme

Corollary 8.2 (Failure of divisor completeness for the cube). *Divisor completeness does not hold for the cube: all $n \geq 1$ admit non-transitive slab dissections. The emerging dichotomy is:*

	<i>Tetrahedron (T_d)</i>	<i>Cube (O_h)</i>
<i>G-transitive: $n \mid G$ sufficient?</i>	✓ (8 divisors)	✓ (10 divisors)
<i>Non-G-transitive constructions</i>	none known	all n (slabs)
<i>Parallel faces</i>	0	3

Hence [conjecture 1.1](#) is *not* a generic polyhedral phenomenon: it is specifically the absence of parallel faces on the regular tetrahedron that prevents non-transitive constructions and motivates [conjecture 1.2](#).

Remark 8.3 (Arithmetic note). The aggregate divisor sums do not transfer:

$$\frac{\sigma(24)}{24} = \frac{60}{24} = \frac{5}{2}, \quad \frac{\sigma(48)}{48} = \frac{124}{48} = \frac{31}{12}.$$

9 The Burnside ring invariant

The Burnside ring $\text{Burn}(T_d)$ [2, 13] provides a representation-theoretic framework for characterising transitive G -sets and detecting obstructions to non-transitive dissections.

9.1 The subgroup lattice of $T_d \cong S_4$

The tetrahedral group $T_d \cong S_4$ has 11 *conjugacy classes of subgroups* and 30 subgroups in total ([Table 7](#)).

Table 7: The 11 conjugacy classes of subgroups of S_4 .

Class	Subgroup H	$ H $	$[S_4 : H]$	# conjugates
0	$\{e\}$	1	24	1
1	$\mathbb{Z}_2^t$ (transposition)	2	12	6
2	$\mathbb{Z}_2^d$ (double transposition)	2	12	3
3	$\mathbb{Z}_3$	3	8	4
4	V_4 (non-normal Klein-4)	4	6	3
5	V_4^n (normal Klein-4)	4	6	1
6	$\mathbb{Z}_4$	4	6	3
7	S_3	6	4	4
8	D_4	8	3	3
9	A_4	12	2	1
10	S_4	24	1	1

The valid orbit sizes for any T_d -action are thus $\{1, 2, 3, 4, 6, 8, 12, 24\}$ – precisely the eight divisors of 24.

9.2 Equivariant Euler characteristic

The barycentric subdivision of T yields a T_d -equivariant CW-complex. A direct count gives

$$\chi(T) = 15 - 50 + 60 - 24 = 1, \tag{5}$$

consistent with T being a contractible 3-ball. Lifting to the Burnside ring:

Computational Certificate 9.1 (Equivariant Euler class). *The equivariant Euler characteristic $\chi_{T_d}(T)$ of T , viewed as an equivariant CW-complex, equals the class of the point: $\chi_{T_d}(T) = [T_d/T_d] \in \text{Burn}(T_d)$, with character vector $(1, 1, \dots, 1) \in \mathbb{Z}^{11}$.*

9.3 The mark table

The mark table $M[i][j] = |\text{Fix}_{H_j}(T_d/H_i)|$ encodes the fixed-point structure of every transitive T_d -set. Its computed value is given in [Table 8](#); it has full rank 11, confirming that $\text{Burn}(T_d)$ is the free abelian group of rank 11 on the transitive T_d -sets.

Table 8: Mark table of $T_d \cong S_4$. Rows indexed by transitive T_d -sets T_d/H_i (top-to-bottom: subgroup order ascending); columns indexed by subgroups H_j (left-to-right: same ordering).

T_d/H_i	$\{e\}$	$\mathbb{Z}_2^t$	$\mathbb{Z}_2^d$	$\mathbb{Z}_3$	V_4	V_4^n	$\mathbb{Z}_4$	S_3	D_4	A_4	S_4
$T_d/\{e\}$	24	0	0	0	0	0	0	0	0	0	0
$T_d/\mathbb{Z}_2^t$	12	2	0	0	0	0	0	0	0	0	0
$T_d/\mathbb{Z}_2^d$	12	0	4	0	0	0	0	0	0	0	0
$T_d/\mathbb{Z}_3$	8	0	0	2	0	0	0	0	0	0	0
T_d/V_4	6	2	2	0	2	0	0	0	0	0	0
T_d/V_4^n	6	0	6	0	0	6	0	0	0	0	0
$T_d/\mathbb{Z}_4$	6	0	2	0	0	0	2	0	0	0	0
T_d/S_3	4	2	0	1	0	0	0	1	0	0	0
T_d/D_4	3	1	3	0	1	3	1	0	1	0	0
T_d/A_4	2	0	2	2	0	2	0	0	0	2	0
T_d/S_4	1	1	1	1	1	1	1	1	1	1	1

9.4 Coset decomposition – algebra matches geometry

For every known dissection the Burnside ring element is a single transitive T_d -set:

Table 9: Each dissection is a single transitive T_d -set in $\text{Burn}(T_d)$.

S_n	Coset space	Stabiliser	$\text{Burn}(T_d)$ -element
S_1	T_d/S_4	S_4	$[T_d/S_4]$
S_2	T_d/A_4	A_4	$[T_d/A_4]$
S_3	T_d/D_4	D_4	$[T_d/D_4]$
S_4	T_d/S_3	S_3	$[T_d/S_3]$
S_6	T_d/V_4	V_4	$[T_d/V_4]$
S_8	$T_d/\mathbb{Z}_3$	$\mathbb{Z}_3$	$[T_d/\mathbb{Z}_3]$
S_{12}	$T_d/\mathbb{Z}_2^t$	$\mathbb{Z}_2^t$	$[T_d/\mathbb{Z}_2^t]$
S_{24}	$T_d/\{e\}$	$\{e\}$	$[T_d/\{e\}]$

9.5 Geometric vs. algebraic stabilisers

We now make precise the relation between the two notions of symmetry attached to the atlas: the piece-type stabiliser of [Definition 2.1](#) (recorded in [Table 4](#), “algebraic”) and the partition-level stabiliser G_{diss} of [Definition 2.2](#) (recorded in [Table 10](#), “geometric”).

Lemma 9.2 (Geometric vs. algebraic stabilisers). *For each S_n in the atlas let P_0 be the canonical piece of [Table 2](#), let $H_n := \text{Stab}_{T_d}(P_0)$ be its piece-type stabiliser ([Definition 2.1](#), $|H_n| = 24/n$), and let $G_{\text{diss}}(S_n)$ be its geometric symmetry group ([Definition 2.2](#)). Then:*

1. $G_{\text{diss}}(S_n)$ is the subgroup of T_d listed in [Table 10](#), and the orbit–stabiliser identity $|G_{\text{diss}}(S_n)| = n \cdot |\text{Stab}_0|$ holds for every n , where $\text{Stab}_0 = G_{\text{diss}}(S_n) \cap H_n$ is the stabiliser of P_0 inside the geometric group.
2. $G_{\text{diss}}(S_n) = T_d$ (i.e. S_n is T_d -invariant in the sense of [Definition 2.3](#)) for $n \in \{1, 12, 24\}$, and only for these n .

3. For $n \in \{2, 3, 4, 6, 8\}$ one has $G_{\text{diss}}(S_n) \subsetneq T_d$; the elements of $T_d \setminus G_{\text{diss}}(S_n)$ send S_n to a distinct but congruent partition of T .
4. In every case S_n is G_{diss} -transitive (Definition 2.4), i.e. the n pieces form a single orbit under $G_{\text{diss}}(S_n)$; this is the statement certified in Computational Certificate 7.1.

Proof. The numerical data of Table 10 are verified by direct computation in the embedding (1): for each S_n we enumerate the 24 elements $g \in T_d$ and test whether the image partition $g(S_n)$ equals S_n setwise; the set of such g is $G_{\text{diss}}(S_n)$. Items (2) and (3) are read off the table. The orbit–stabiliser identity in (1) is the usual one for the transitive action of $G_{\text{diss}}(S_n)$ on the n pieces of S_n ; transitivity within $G_{\text{diss}}(S_n)$ is verified case by case in the supplementary certificate. Item (4) is a restatement of the definitions once transitivity of the G_{diss} -action is established. $\square$

Remark 9.3 (Summary). For $n \in \{1, 12, 24\}$ all four notions of Section 2 coincide; for $n \in \{2, 3, 4, 6, 8\}$ only G_{diss} -transitivity (Definition 2.4) holds, and $G_{\text{diss}}(S_n) \subsetneq T_d$ is strictly smaller than T_d . In particular the Lagrange obstruction $n \mid 24$ (Corollary 7.4) uses only G_{diss} -transitivity.

Table 10: Geometric vs. algebraic stabilisers of the eight dissections.

S_n	G_{diss}	$ G_{\text{diss}} $	Transitive?	$ \text{Stab}_0 $
S_1	S_4	24	✓	24
S_2	V_4	4	✓	2
S_3	S_3	6	✓	2
S_4	D_4	8	✓	2
S_6	S_3	6	✓	1
S_8	D_4	8	✓	1
S_{12}	S_4	24	✓	2
S_{24}	S_4	24	✓	1

10 IFS dissection grammar

Every halving refinement in the poset of Sections 5 and 6 can be encoded as an iterated function system (IFS): a pair of affine contractions whose images tile the parent piece. This produces a formal grammar for generating all eight dissections.

10.1 Halving rules

Table 11: Halving rules and their chirality. Rule R3 ($S_4 \rightarrow S_8$) is the unique rule with *both* branches orientation-preserving.

Rule	Production	Geometric name	$\det(f_1)$	$\det(f_2)$
R1	$S_1 \rightarrow f_1(S_2) \cup f_2(S_2)$	Median bisection	+1	−1
R2	$S_2 \rightarrow g_1(S_4) \cup g_2(S_4)$	Orthogonal median	+1	−1
R3	$S_4 \rightarrow h_1(S_8) \cup h_2(S_8)$	Longest-edge bisection	+1	+1
R4	$S_3 \rightarrow p_1(S_6) \cup p_2(S_6)$	Axial median bisection	+1	−1
R5	$S_{12} \rightarrow q_1(S_{24}) \cup q_2(S_{24})$	Edge bisection	+1	−1

10.2 Heterogeneous bridge (R6)

The heterogeneous bridge (theorem 6.1) corresponds to the rule R6: $S_2 \rightarrow \alpha(S_3) \cup \beta(S_6)$ with $\det(\alpha)\det(\beta) = -1$.

10.3 Formal alphabet and grammar

The alphabet of the grammar is $\Sigma = \{S_1, S_2, S_3, S_4, S_6, S_8, S_{12}, S_{24}\}$. Homogeneous productions are R1–R5; the unique heterogeneous production is R6. Derived multi-level compositions include $R1 \circ R2$ ($S_1 \rightarrow 4 \times S_4$, quadrisection) and $R1 \circ R2 \circ R3$ ($S_1 \rightarrow 8 \times S_8$, 8T-LE octasection).

10.4 Composition $S_1 \rightarrow 8 \cdot S_8$ and PPS23 similarity classes

Iterated longest-edge bisection of T produces a finite sequence of similarity classes [11]; our grammar reproduces and extends their enumeration:

Table 12: Similarity class count under iterated longest-edge bisection.

Depth	Pieces	Similarity classes
1	2	1
2	4	2
3	8	4
4	16	8
5	32	9
6	64	12
7	128	16
8	256	17
9	512	18

The class count stabilises at 18 by depth 9, consistent with a finite attractor in similarity space. At depth 9 the four dominant classes are the regular tetrahedron ($\times 40$), an isosceles variant ($\times 40$), the S_8 -type ($\times 108$) and the S_3 -type ($\times 40$).

11 An obstruction stack for T_d -invariant dissections (screening)

Status of this section (honesty block). This section does *not* resolve the non- G_{diss} -transitive case of [conjecture 1.1](#). It builds an *evidence stack* that, under the standing hypothesis of T_d -invariance ([Definition 2.3](#)) and a T_d -maximality argument flagged as *plausible but not fully rigorous* ([Section 11.3](#) and [remark 11.4](#)), filters every tested orbit decomposition for $n \leq 16$, $n \nmid 24$. Of the nine such values of n four pass through the crystallographic restriction alone (*unconditional* screening in the T_d -invariant category) and five additionally invoke the T_d -maximality step, tagged ($\dagger$) in [Table 13](#) and recorded at level L1+ in [Table 14](#). A theorem-level resolution of the non-transitive problem, without the T_d -maximality caveat, is the content of the Stabiliser Transport Lemma ([Lemma 12.2](#), [Section 12](#)) and is not achieved here.

The remaining question for [conjecture 1.1](#) is whether a congruent n -dissection of T with $n \nmid 24$ could exist without being G_{diss} -transitive ([Definition 2.4](#)). We develop a systematic obstruction framework combining the subgroup lattice, the crystallographic restriction theorem, and a T_d -maximality argument; the resulting stack filters all tested decompositions in the T_d -invariant category ([Definition 2.3](#)). It is important to stress that T_d -invariance is strictly stronger than mere non-transitivity under G_{diss} : a hypothetical congruent dissection might have $G_{\text{diss}}(D) \subsetneq T_d$ and therefore escape the T_d -orbit analysis of this section entirely.

11.1 Orbit decomposition

Suppose a congruent n -dissection $T = \bigsqcup_{i=1}^n P_i$ is T_d -invariant ([Definition 2.3](#)), i.e. every $\sigma \in T_d$ permutes the pieces (equivalently, $\sigma(\{P_1, \dots, P_n\}) = \{P_1, \dots, P_n\}$ for all σ). All eight known dissections enjoy this property for $n \in \{1, 12, 24\}$ ([lemma 9.2](#)); for $n \in \{2, 3, 4, 6, 8\}$ only the smaller group $G_{\text{diss}}(S_n) \subsetneq T_d$ preserves the partition set-wise and the analysis below strictly

speaking applies only after replacing T_d by $G_{\text{diss}}(S_n)$. The present section enforces T_d -invariance as a standing hypothesis so as to have the full group T_d available for the orbit decomposition. Under this assumption, the T_d -action decomposes into orbits $\mathcal{O}_1, \dots, \mathcal{O}_k$ with $|\mathcal{O}_j| = [T_d : H_j]$ for stabiliser $H_j \leq T_d$. Since each orbit size divides $|T_d| = 24$:

$$n = \sum_{j=1}^k [T_d : H_j], \quad [T_d : H_j] \in \{1, 2, 3, 4, 6, 8, 12, 24\}. \quad (6)$$

For $n \nmid 24$ necessarily $k \geq 2$.

Lemma 11.1 (Size-1 orbit). *At most one orbit has size 1.*

Proof. A T_d -invariant piece must contain the centroid G_T , the unique T_d -fixed point of T . Two distinct convex pieces containing G_T would overlap. $\square$

11.2 Cross-orbit symmetry obstruction

Let $P_1 \in \mathcal{O}_1$ and $P_2 \in \mathcal{O}_2$ be pieces in distinct orbits of a congruent T_d -invariant dissection. Since $P_1 \cong P_2$, a fixed congruence identifies them with a common abstract reference tile P whose intrinsic symmetry group is $K := \text{Sym}(P) \leq O(3)$. The stabilisers

$$H_1 = \text{Stab}_{T_d}(P_1), \quad H_2 = \text{Stab}_{T_d}(P_2)$$

act on P and hence embed (as subgroups, up to conjugation in $O(3)$) into K .

Lemma 11.2 (Stabiliser pair screening). *Let $H_1, H_2 \leq T_d$ be two point-stabiliser conjugacy classes from the 30 subgroups of T_d (in 11 conjugacy classes). Let $K(H_1, H_2) \leq O(3)$ denote the smallest finite subgroup of $O(3)$, up to conjugation, that contains a pair of non-conjugate copies of H_1 and H_2 (whenever such a pair exists). Then, using the classical classification of finite subgroups of $O(3)$ and the crystallographic restriction theorem [9], the following non-conjugate stabiliser pairs force K to contain a full cubic group $\geq S_4$:*

Non-conjugate pair (H_1, H_2)	Minimal containing $K \leq O(3)$
$(D_4, \mathbb{Z}_3)$	$\supseteq S_4$ (mixed 4-fold & 3-fold)
$(\mathbb{Z}_4, \mathbb{Z}_3)$	$\supseteq S_4$ (same mechanism)
(S_3, V_4)	$\supseteq S_4$ (3-fold + orthogonal 2-folds)
(A_4, D_4)	$\supseteq S_4$ (automatic)
(A_4, S_3)	$\supseteq S_4$ (automatic)
(D_4, S_3)	$\supseteq S_4$ (automatic)

In particular, for each such pair, any abstract tile P admitting both H_1 and H_2 as non-conjugate subgroups of $\text{Sym}(P)$ satisfies $|\text{Sym}(P)| \geq 24$ and $\text{Sym}(P)$ contains a copy of S_4 acting on a cube.

Proof sketch. Each listed pair either (i) combines a 4-fold and a 3-fold rotation, or (ii) combines a 3-fold rotation with a V_4 subgroup of orthogonal 2-folds, or (iii) already contains A_4 . In case (i) the crystallographic restriction and the classification of finite rotation groups of $\mathbb{R}^3$ force the generated subgroup to be the rotation group of the cube (order 24) or its full symmetry group (order 48); in case (ii) the three mutually orthogonal 2-folds and the 3-fold permuting them generate the same rotation subgroup; case (iii) is immediate since $A_4 \leq S_4$. A finite verification on all unordered pairs of conjugacy classes in T_d (performed computationally; see `independent_verifier.py` in the supplementary code capsule) confirms no other non-conjugate pair realises a smaller containing group. $\square$

Remark 11.3 (Why this is a screening lemma, not an impossibility theorem). [lemma 11.2](#) is used as a *finite screening device* on orbit decompositions of $n \nmid 24$: whenever a putative decomposition assigns stabilisers in a pair of the table above to distinct orbits, the abstract tile must admit a cubic-or-larger symmetry group, which is incompatible with the finite list of tile shapes compatible with a tetrahedral dissection (verified case-by-case in the obstruction stack, [Table 13](#)). It is *not* a standalone impossibility theorem; the final kill still requires either T_d -maximality ([Section 11.3](#), with the caveat of [remark 11.4](#)) or the Stabiliser Transport Lemma ([lemma 12.2](#)).

11.3 The T_d -maximality argument

Status. The argument of this subsection is **plausible but not fully rigorous**: it relies on the implicit assumption that the cross-orbit congruence $P_1 \rightarrow P_2$ extends to a global isometry of T , which is not forced by the dissection hypothesis alone. See [remark 11.4](#) immediately below and the Stabiliser Transport Lemma ([lemma 12.2](#), [Section 12](#)) for the correct group-theoretic completion. Decompositions marked $(\dagger)$ in [Table 13](#) are precisely those whose filtering depends on this conditional argument.

For orbit decompositions surviving the crystallographic filter – those with *same-type* stabilisers (e.g. $n = 9 = 3 + 3 + 3$ with all D_4) – the cross-orbit congruence map $P_1 \rightarrow P_2$ must be an isometry of $\mathbb{R}^3$ that is *not* in T_d : T_d permutes pieces within orbits, not between them. Since $T_d = \text{Isom}(T)$ is the full isometry group of T , any such map cannot preserve T globally.

Remark 11.4 (Caveat on [Section 11.3](#)). A congruence $\sigma: P_i \rightarrow P_j$ only needs to map one piece to another; it does not need to preserve T globally. Hence $\sigma \notin T_d$ does not *itself* produce a contradiction: the argument of [Section 11.3](#) is plausible but not fully rigorous, and the Stabiliser Transport Lemma of [Section 12](#) ([lemma 12.2](#)) provides the correct group-theoretic completion.

11.4 Results of the obstruction stack

Systematic enumeration of all integer partitions of n into valid orbit sizes ([\(6\)](#)), followed by stabiliser-type analysis, gives:

Table 13: Non-transitive orbit decompositions filtered for $n \leq 16$, $n \nmid 24$. A dagger $(\dagger)$ indicates decompositions that require the T_d -maximality argument of [Section 11.3](#), plausible but not yet rigorous.

n	Total decomp.	After filter	All filtered?	Primary mechanism
5	6	1	yes	$K \geq S_4$
7	12	3	yes	$K \geq S_4$
9	22	6	yes †	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
10	30	8	yes †	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
11	36	10	yes	$K \geq S_4$
13	58	17	yes	$K \geq S_4$
14	76	21	yes †	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
15	90	26	yes †	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
16	115	33	yes †	$K \geq S_4 + \text{same-type} + T_d\text{-max}$

Four of the nine cases are rigorously filtered by the crystallographic restriction alone (marked without dagger in [Table 13](#)); the remaining five additionally require the T_d -maximality argument ([Section 11.3](#)), which is plausible but not yet fully rigorous – see [remark 11.4](#). We therefore do *not* claim theorem-level impossibility of non- G_{diss} -transitive dissections for $n \leq 16$: what the table establishes is that, within the T_d -invariant category, every tested orbit decomposition is ruled out *up to* the T_d -maximality caveat. This is recorded at level L1+ in [Table 14](#) and motivates (but does not prove) [conjecture 1.2](#).

11.5 Argument chain summary

Table 14: Evidence-level stack of the screening argument against non- G_{diss} -transitive dissections, under the standing hypothesis of T_d -invariance.

Step	Statement	Status
1	Non-transitive $\Rightarrow \geq 2$ orbits, sizes divide 24	Theorem
2	At most one T_d -invariant piece (lemma 11.1)	Theorem
3	Stabiliser action on vertices forces $ V(P) \cap V(T) \in \{0, 4\}$	Proposition (convex)
4	Mixed stabiliser types $\Rightarrow K \geq S_4$ (crystallographic restriction)	Theorem
5	Same-type orbits require non- T_d isometry	Plausible (L1+)
6	S_4 -symmetric bodies cannot tile T for $n \nmid 24$	L1 evidence

12 The rigidity and configurational programme

We now describe the programme that attacks [conjecture 1.2](#) and contains the two principal new impossibility theorems of the paper: [conditional theorem 12.16](#) ($n = 5$ conditional reduction) and [conditional theorem 12.8](#) ($n = 16$ all-one-face conditional reduction).

Throughout this section we adopt two standing hypotheses, consistent with all prior literature on congruent simplicial tilings: (i) every piece is convex, and (ii) the dissection is *face-to-face* (every interior 2-face of the piece complex is shared by exactly two pieces).

12.1 Dihedral rigidity programme

A direct attack via dihedral-angle analysis exploits the irrationality of α/π (Conway–Jones [3]). The intended chain was: $\alpha/\pi \notin \mathbb{Q} \Rightarrow$ interior edges carry zero α -coefficient $\Rightarrow$ Dehn localises to the original T -edges $\Rightarrow$ every piece is an α -carrier $\Rightarrow$ every piece touches ≥ 2 of T 's 4 faces $\Rightarrow$ Face-Normal Rigidity forces the gluing isometry $\phi \in T_d \Rightarrow$ transitivity $\Rightarrow n \mid 24$.

Theorem 12.1 (Face-Normal Rigidity). *Let $Q \in O(3)$ be an orthogonal linear map. If Q sends at least two of the four face normals of T (as unit vectors based at the origin) to face normals of T , then Q is the linear part of some element of T_d . Equivalently, for every isometry $\phi(x) = Qx + t$ of $\mathbb{R}^3$ whose linear part Q sends two or more face normals of T to face normals of T , one has $Q \in T_d$ (regarded as a subgroup of $O(3)$ in the standard way). Verified exhaustively on all 144 ordered pairs of face normals.*

However the chain above is broken by two identified gaps:

GAP A (Dehn localisation). The chain assumes all piece dihedral angles have the form $\theta = m\alpha + q\pi$ with $m \in \mathbb{Z}$. This is unjustified for arbitrary cuts: only original T -edges carry α -related angles a priori, and global cancellation $\sum m_i = 0$ at an interior edge does *not* force each m_i to vanish. The naïve “ $m \geq 2 \Rightarrow \theta > \alpha$ ” dichotomy also fails, e.g. $\theta = 2\alpha - \pi/2 \approx 51^\circ < \alpha$.

GAP B (Intrinsic boundary). A face of a piece lying on ∂T is not intrinsically distinguishable from an interior face: the gluing isometry $\phi: P_1 \rightarrow P_2$ could map a boundary face of P_1 to an interior face of P_2 . Without intrinsic distinguishability the Face-Normal Rigidity theorem ([theorem 12.1](#), which *is* valid as a finite test) cannot be invoked to force $\phi \in T_d$.

[theorem 12.1](#) is therefore a correct rigidity tool, but its deployment requires first discharging GAPs A and B. The remainder of [Section 12](#) addresses these gaps.

12.2 Reduction engine and Stabiliser Transport

Following the GAP A/B diagnosis we pivot to a *reduction strategy*: defer the final kill and restrict the shape of any putative counterexample.

Lemma 12.2 (Stabiliser Transport – L2 theorem). *Let $T = P_1 \cup \dots \cup P_n$ be a congruent dissection with congruence group $G = \langle \sigma_{ij} \rangle \leq \text{Isom}(\mathbb{R}^3)$. Then:*

- (a) *stabilisers of congruent pieces are conjugate in $\text{Isom}(\mathbb{R}^3)$;*
- (b) *if $n \nmid 24$ then $G \not\leq T_d$; consequently G contains a non-tetrahedral congruence and the T_d -action on the pieces has ≥ 2 orbits.*

lemma 12.2(a) is standard; (b) follows from the fact that T_d has 30 subgroups in 11 conjugacy classes with index set $\{24/|H|\} = \text{div}(24)$.

Definition 12.3 (Boundary-Normal set). For each face f of the abstract tile P , the *boundary-normal set*

$$\text{BN}(f) := \{ i \in \{0, 1, 2, 3\} : \exists \text{ placement } \phi \text{ with } \phi(f) \subset F_i(T) \}$$

records which T -faces $F_0, \dots, F_3$ the abstract face f can land on.

Conjecture 12.4 (Boundary-Normal Orbits, L1). In a minimal counterexample to [conjecture 1.2](#), the boundary faces of the abstract tile P fall into at most two $\text{Sym}(P)$ -orbits under the 4-normal dictionary $\text{BN}(\cdot)$.

The face-orbit analysis for all 11 subgroup conjugacy classes of T_d partitions $\{0, 1, 2, 3\}$ into progressively finer schemes: e.g. $\mathbb{Z}_3$ acts as $\{0\} \mid \{1, 2, 3\}$, V_4 as $\{0, 1\} \mid \{2, 3\}$, which constrains the boundary structure of any counterexample.

12.3 Spherical Link Program and the Conditional Separation Lemma

The local geometry at each T -vertex V is captured by the *vertex link*, an equilateral spherical triangle Δ_V . Its angles are equal to the dihedral α of T , and its sides equal the face-angle $\pi/3$: both are exactly computable.

Proposition 12.5 (Vertex-link geometry, L2). *The vertex link Δ_V at every vertex of T is equilateral with spherical angle $\alpha = \arccos(1/3)$ and side length $\pi/3$.*

Proof. Apply the spherical law of cosines to the three faces meeting at V : $\cos a = \cos \alpha / (1 - \cos \alpha) = 1/2$, whence $a = \pi/3$. Equality of the three angles is immediate from the symmetry $\text{Sym}_V(T)$. $\square$

Proposition 12.6 (Conditional Separation Lemma, L1+). *Suppose Hypothesis H holds: every interior angle of every link-tile at every T -vertex belongs to $\mathbb{Q}\alpha + \mathbb{Q}\pi$. Then under the Gauss–Bonnet constraint on s -gon tiles Q of Δ_V , the identity separates into*

$$\sum_{\ell} a_{\ell} = \frac{3}{k}, \quad \sum_{\ell} b_{\ell} = s - 2 - \frac{1}{k}, \quad a_{\ell}, b_{\ell} \in \mathbb{Q},$$

where k is the number of pieces meeting at V . In a uniform convex model this forces $k \in \{1, 3, 6, 9, 12, \dots\}$, eliminating $k \in \{2, 4, 5\}$.

Three open gaps prevent [proposition 12.6](#) from attaining theorem status: **GAP C** (Hypothesis H), **GAP D** (mixed vertex types may produce non-congruent link-tiles), **GAP E** (non-convex pieces). Discharge of GAP C is the key conditional step.

12.4 Configurational closure at $n = 16$: the all-one-face residual

[conjecture 12.4](#) targets GAP B by classifying boundary-face orbits abstractly. The same gap can be attacked *configurationally*: enumerate, for fixed piece count n , the candidate “abstract role distributions” of pieces on ∂T and eliminate them one by one.

For $n = 16$ the most constrained distribution places one boundary face of every piece on the *same* single T -face; we call this the *all-one-face* residual.

Setup. Embed T via the rational coordinates

$$v_0 = (1, 1, 1), \quad v_1 = (1, -1, -1), \quad v_2 = (-1, 1, -1), \quad v_3 = (-1, -1, 1), \quad (7)$$

so $|v_i v_j|^2 = 8$ and $V_T = 8/3$. The medial triangle of each T -face decomposes that face into four sub-triangles (*slots*): three corner slots (at the three T -vertices) and one central slot (the medial triangle itself). Two auxiliary hypotheses derived from the preparatory MILP (see the P25-C7 slot decomposition chain of the supplementary materials) cut the search space:

(H_{roles}) (Role-type assignment). The medial decomposition is the only abstract role-type assignment compatible with face-to-face adjacency: each boundary footprint is either a corner slot or the central slot.

(H_{slot}) (Slot separation). Every piece’s boundary footprint is exactly one of the four slots above; no piece’s footprint mixes slot types.

Both (H_{roles}) and (H_{slot}) are unconditional in the *Coxeter-pure* instance (where every piece is a union of A_3 -orthoschemes) but residual in the general convex instance.

Theorem 12.7 ($n = 16$ all-one-face: Coxeter-pure instance). *There is no face-to-face Coxeter-pure congruent dissection of T into 16 mutually congruent pieces such that every piece has exactly one boundary face on a single common T -face.*

Proof. In the Coxeter-pure instance both (H_{roles}) and (H_{slot}) are automatic (a union of A_3 -orthoschemes has a unique role assignment and a unique boundary-slot type per piece; see [Section 12.4](#)). The conclusion therefore follows from the conditional [conditional theorem 12.8](#) below. $\square$

Conditional Theorem 12.8 ($n = 16$ all-one-face: general convex instance). *Assume (H_{roles}) + (H_{slot}). Let $\{P_1, \dots, P_{16}\}$ be a face-to-face congruent dissection of T into 16 mutually congruent convex polytopes such that every P_i has exactly one boundary face on a single common T -face. Then no such dissection exists.*

The proof assembles three rigidity lemmas, each independently certified by exact rational arithmetic (Python **Fractions**, zero failed tests across the chain).

Lemma 12.9 (Tetrahedrality reduction, C24x). *Every P_i is a tetrahedron with triangular medial base on ∂T and a single interior apex.*

Proof. Two parts.

(I) *Scaled-top family.* No *scaled-top* piece – right prism ($t = 1$), right frustum ($t \in (0, 1)$), or right pyramid with apex above the base centroid ($t = 0$) – fits. The volume constraint $V_{\text{piece}} = V_T/16 = 1/6$ with base area $\sqrt{3}/2$ forces height $h_{\text{piece}} = h_T/4$ (prism) or the frustum-corrected analogue.

- *Layer A (prism).* At $h_{\text{prism}} = \sqrt{3}/9$ the T -cross-section has squared inradius $(121/144)(2/3)$, while the medial triangle has squared circumradius $2/3$; the ratio $144/121 > 1$, so at least one medial vertex sticks out of T .

- *Layer B (frustum)*. For any base point Q the difference between the corner-frustum top and the central-frustum top equals $(1-t)(c_{\text{corner}} - c_{\text{central}})$. Since $|c_{\text{corner}} - c_{\text{central}}|^2 = 2/3 \neq 0$ exactly, face-to-face matching on the shared medial edge fails for every $t \in [0, 1)$.

(II) *Multi-apex residual*. Every convex piece with $k \geq 2$ interior vertices is excluded by two independent rigorous arguments. Let F_1 be the lateral face of a corner- v_j piece carrying the medial edge (m_{jk}, m_{jl}) , and let q be the number of interior vertices of the piece on F_1 .

- **(ARG-1) F_4 parity-pairing** ($k = 2, q = 2$, Type-2 5-vertex polytope). The apex-interior face $F_4 = (v_j, A, B)$ contains v_j and both interior vertices but no midpoint. At vertex v_j exactly three corner- v_j pieces meet, each carrying one such F_4 . Face-to-face matching forces these three faces to pair *among themselves*, but a fixed-point-free involution on a 3-element set is impossible.
- **(ARG-2) Three-plane collapse** (all $k \geq 2, q \geq 2$). Congruence on edge multisets forces every interior vertex X on F_1 of a corner- v_1 piece to satisfy $|v_1 - X|^2 = |m_{kl} - X|^2$, i.e. X lies on the perpendicular bisector plane of (v_1, m_{kl}) . Face-to-face propagation across the three T -faces incident to v_1 yields three such planes:

$$2x - y - z = 1, \quad x - y - 2z = 1, \quad x - 2y - z = 1.$$

The coefficient matrix is non-singular with determinant -4 ; the unique intersection point is

$$P_* = \frac{1}{4}(1, -1, -1),$$

a rational interior point of T with barycentric coordinates $(3, 7, 3, 3)/16$. Every interior vertex on F_1 collapses to P_* , contradicting $q \geq 2$ distinct vertices.

- The case $q = 1$ (triangular F_1) reduces to [lemma 12.10](#) below via face-by-face apex collapse along the three medial edges.

This completes the proof. □

Lemma 12.10 (Apex collapse, C24w). *Conditional on [lemma 12.9](#), every congruent piece has a single interior apex at perpendicular height $h_T/4$ above its medial base, and four rigidity layers force all 16 apices to coincide at the centroid $G_T = (0, 0, 0)$:*

- (L_1) *Volume forces apex height $= h_T/4$.*
- (L_2) *Same-face medial-edge matching forces the apex of the corner- v_j piece on face F_i to equal the apex of the central piece on F_i .*
- (L_3) *T -edge matching propagates the apex across faces, forcing all four central apices to coincide at a common point Q .*
- (L_4) *Q is at perpendicular distance $h_T/4$ from every face; the 3-of-4 face-normal linear system has unique solution $Q = G_T$.*

Lemma 12.11 (Edge-multiset incongruence at G_T , C24v). *At $Q = G_T$, the corner piece has squared-edge multiset $\{1, 1, 2, 2, 2, 3\}$ and the central piece has squared-edge multiset $\{1, 1, 1, 2, 2, 2\}$. These multisets differ; hence the corner and central pieces are not congruent.*

Proof of conditional theorem 12.8. Combine [lemma 12.9](#), [lemma 12.10](#), and [lemma 12.11](#) with the medial-base reduction encoded in $(H_{\text{roles}}) + (H_{\text{slot}})$: the first forces tetrahedrality; the second collapses all apices to G_T ; the third produces a direct violation of piece congruence. □

Remark 12.12 (Scope of [theorem 12.7](#) and [conditional theorem 12.8](#)). In the Coxeter-pure instance $(H_{\text{roles}}) + (H_{\text{slot}})$ are automatic, so [theorem 12.7](#) is the L3 (unconditional) specialisation of the L2 conditional statement [conditional theorem 12.8](#). In the general convex instance $(H_{\text{roles}}) + (H_{\text{slot}})$ are residual; their theorem-level discharge is the remaining isolated bridge lemma (the paired Star^+ transport lemma of the supplementary programme), so [conditional theorem 12.8](#) is at evidence level L2.

Remark 12.13 (Convexity hypothesis). The convexity of the pieces is essential to [lemma 12.9](#): without it, multi-apex non-convex pieces are not reached by the perpendicular-bisector argument (ARG-2). Discharging convexity is natural future work; the present theorem is, as stated, in the convex category.

Remark 12.14 (Why $n = 16$?). $n = 16$ divides neither 24 nor 48. Note that closing the full $n = 16$ case *would* advance [conjecture 1.1](#), since $16 \nmid 24$ means that the conjecture predicts no dissection should exist. However, [lemma 12.9](#) closes only the *all-one-face residual* configuration at $n = 16$, not the entire case. This particular configuration was historically the most resistant residual of the role-distribution table in the programme described in [Section 12.2](#). The same C24 machinery is structurally portable to the other non-divisors $n \in \{20, 28, \dots\}$ and gives a template for the configurational attack on GAP B.

12.5 Conditional reduction at $n = 5$: the first non-divisor reduced to finite hypotheses

The C24-chain of [Section 12.4](#) reduced the largest role-distribution residual at $n = 16$ to a conditional statement. We now reduce the smallest non-divisor of 24, namely $n = 5$, in the convex category, modulo four explicit finite hypotheses: (H-trip) (topological; heuristic input to (H-coc), not used directly in the formal proof of [Conditional Theorem 12.16](#); see [Remark 12.29](#)), (H-coc) (triangle cocycle on orbit-transport bits, used in the single $(12, 12, 24)/(2, 2)$ -orbit sub-case of the $f = 7$ branch; [Lemma 12.28](#)), (H-orb) (orbit detection, used wherever the orbit-transport [Lemma 12.26](#) is invoked; [Remark 12.27](#)), and (H-Qb) (single-piece Qb exclusion in $f = 5$; [Remark 12.31](#)). Four of the seven admissible (t, k) -residuals of the role-distribution census are closed *unconditionally*; the remaining three invoke (H-orb) and (H-Qb) uniformly, and (H-coc) is additionally used on the single $(12, 12, 24)/(2, 2)$ -orbit $f = 7$ sub-case. Throughout this subsection the standing embedding is (7).

Status of the four hypotheses. We emphasise up front that [Conditional Theorem 12.16](#) is a *conditional reduction*, not an unconditional closure of $n = 5$:

- (H-trip) (*topological; heuristic*). Not used directly in the formal proof of [Conditional Theorem 12.16](#); its intended role is as the input to a developing-map reduction $(\text{H-trip}) \Rightarrow (\text{H-coc})$ sketched but *not completed* in [Remark 12.29](#).
- (H-coc) (*algebraic; required, standing*). A triangle cocycle in $\text{Sym}(P)$ on the K_5 adjacency graph; used in exactly one sub-case of the $f = 7$ branch. A concrete finite certificate on any candidate dissection is included in the data and code capsule accompanying the submission (see the availability statement in [Section 1](#)).
- (H-orb) (*algebraic; required, standing*). Facet orbit-detection on the reference tile; a single finite check per orbit type (see [Remark 12.27](#)).
- (H-Qb) (*geometric; required, standing*). Single-piece exclusion of the Qb-adj/Qb-opp quadrilateral-pyramid sub-configurations in $f = 5$ ([Remark 12.31](#)).

The unconditional closure of $n = 5$ would require either (i) completing the developing-map argument that promotes (H-trip) to (H-coc), or (ii) discharging (H-orb) and (H-Qb) by a direct

finite verification. Both are natural next steps of the programme and are outside the scope of the present paper.

Volume, face area, height of T and per-piece totals in the $n = 5$ case:

$$V_T = \frac{8}{3}, \quad A_T = 2\sqrt{3}, \quad h_T = \frac{4}{\sqrt{3}}, \quad V_{\text{piece}} = \frac{8}{15}, \quad E = \frac{8\sqrt{3}}{5},$$

where E is the per-piece total exposed (boundary) surface area, determined by the constraint $5E = 4A_T =$ total boundary area of T .

Definition 12.15 (Touch count and incidence vector). For each piece P let $I(P) \subseteq \{0, 1, 2, 3\}$ be the set of T -faces on which P has positive 2-area, $t := |I(P)|$ (constant by congruence), and $k = (k_0, k_1, k_2, k_3)$ with $k_j := \#\{P : j \in I(P)\}$, so $\sum_j k_j = 5t$.

Conditional Theorem 12.16 ($n = 5$ convex conditional reduction, modulo (H-trip), (H-coc), (H-orb), (H-Qb)). *Assume the four standing hypotheses (H-trip) (topological; every triple of pieces of a putative 5-piece dissection with K_5 interior adjacency meets in a non-empty triple intersection), (H-coc) (triangle cocycle in $\text{Sym}(P)$; [Lemma 12.28](#)), (H-orb) (orbit detection; [Remark 12.27](#)) and (H-Qb) (single-piece Qb exclusion; [Remark 12.31](#)) for every putative face-to-face convex congruent dissection of the regular tetrahedron T into 5 pieces. Then there is no face-to-face congruent dissection of T into 5 convex pieces. The localisation of each hypothesis in the proof is:*

- (H-trip) enters the proof only via its heuristic reduction to (H-coc) ([Remark 12.29](#)); the logical chain used in the UNSAT arguments invokes (H-coc) directly, so (H-trip) is formally redundant whenever (H-coc) is assumed, and is retained as a standing hypothesis to keep the topological motivation for (H-coc) on the record;
- (H-coc) is invoked only in the $(t = 3, f = 7)$ sub-case with boundary multiset $(12, 12, 24)$ and orbit structure $(2, 2)$ (via [Lemma 12.28](#));
- (H-orb) is invoked wherever the orbit-transport [Lemma 12.26](#) is applied, which is the whole $f = 7$ branch of the three $t = 3$ residuals that go through parity reduction (the weak [Lemma 12.24](#) suffices for $f = 5$ and for every other sub-case);
- (H-Qb) is invoked only in the Qb-adj and Qb-opp sub-cases of the $f = 5$ closure, which appear uniformly in all three parity-reduction residuals.

Four of the seven (t, k) -residuals (both $t = 2$ residuals, the $t = 3$ residual $(5, 5, 3, 2)$, and the $t = 4$ residual) are closed unconditionally (none of the four standing hypotheses is invoked). The remaining three $t = 3$ residuals invoke (H-orb) and (H-Qb) uniformly, and (H-coc) only on the single $(12, 12, 24)/(2, 2)$ -orbit $f = 7$ sub-case.

The proof occupies the remainder of [Section 12.5](#). It proceeds by a role-distribution census (Step 1 below) followed by three UNSAT arguments, one for each admissible value of t (Steps 2 and 3 below).

Step 1: role-distribution census

Since each T -face must be covered by the boundary of the dissection, $k_j \geq 1$ for every j . Each piece contributes area at most E to any single T -face, so $A_T \leq k_j \cdot E$, i.e.

$$k_j \geq \left\lceil \frac{A_T}{E} \right\rceil = \left\lceil \frac{5}{4} \right\rceil = 2. \quad (8)$$

Combined with $\sum_j k_j = 5t$ the lower bound (8) forces $5t \geq 8$, hence $t \geq 2$ (and $t \leq 4$ trivially). In particular, the $n = 5$ analogue of the $n = 16$ all-one-face residual $(t = 1, k = (5, 0, 0, 0))$ is *combinatorially* impossible for two independent reasons: three T -faces would be uncovered (contradicting ∂T -coverage), and $5E = 8\sqrt{3} = 4A_T$, four times the single-face capacity.

After quotienting by S_4 acting on T -faces exactly *seven* residuals survive the filter (8):

t	k -orbit representative (sorted desc)	$\sum k_j$	Per- T -face uniform-share a^2
2	(4, 2, 2, 2)	10	3/4, 3, 3, 3
2	(3, 3, 2, 2)	10	4/3, 4/3, 3, 3
3	(5, 5, 3, 2)	15	12/25, 12/25, 4/3, 3
3	(5, 4, 4, 2)	15	12/25, 3/4, 3/4, 3
3	(5, 4, 3, 3)	15	12/25, 3/4, 4/3, 4/3
3	(4, 4, 4, 3)	15	3/4, 3/4, 3/4, 4/3
4	(5, 5, 5, 5)	20	12/25, 12/25, 12/25, 12/25

Step 2: the $t = 4$ residual (5, 5, 5, 5)

In this residual every piece touches all four T -faces ($t = 4$). Write (m_1, m_2, m_3, m_4) for the canonical boundary-area multiset of the common piece shape; by congruence $M := \sum_j m_j$ is the same for every piece and $5M = 4A_T$, so $M = 4A_T/5$. We make *no* assumption that the m_j are equal.

Lemma 12.17 (Regular-tetrahedron barycentric distance sum). *For any point $p \in T$,*

$$\sum_{j=0}^3 d(p, \text{aff}(F_j)) = h_T,$$

where $F_0, \dots, F_3$ are the four faces of T .

Proof. Standard application of $V_T = (1/3) \sum_j \text{area}(F_j) \cdot d(p, \text{aff}(F_j))$ with all $\text{area}(F_j) = A_T$ and $V_T = (1/3)A_T h_T$. $\square$

Lemma 12.18 (Centroid pyramid decomposition on the closure). *For any convex polytope $P \subset \mathbb{R}^3$ and any point $q \in P$ (closure, not just interior),*

$$\text{Vol}(P) = \frac{1}{3} \sum_{F \in \mathcal{F}(P)} \text{area}(F) d(q, \text{aff}(F)).$$

Every summand is nonnegative; it vanishes if and only if $q \in \text{aff}(F)$.

Proof. Apply the divergence theorem to $v(x) = (x - q)/3$ on P . Since $\text{div } v = 1$,

$$\text{Vol}(P) = \int_P 1 \, dx = \int_{\partial P} v \cdot \hat{n} \, dA = \frac{1}{3} \sum_{F \in \mathcal{F}(P)} \text{area}(F) d_{\text{signed}}(q, \text{aff}(F)),$$

where d_{signed} is the signed distance (positive on the side of the outward normal). For $q \in P$ (convex), every supporting hyperplane has q on its inner side, so all signed distances are nonnegative and equal to the ordinary distance. $\square$

Let q_T be the incenter of T , the unique interior point with $d(q_T, \text{aff}(F_j)) = h_T/4$ for $j = 0, \dots, 3$. Since the five closed pieces cover T and $q_T \in \text{int}(T)$, at least one piece P^* satisfies $q_T \in P^*$. Apply lemma 12.18 at $q = q_T$:

$$V_{\text{piece}} = \underbrace{\frac{1}{3} \sum_{j=0}^3 m_{\sigma^*(j)} d(q_T, F_j)}_B + \underbrace{\frac{1}{3} \sum_{G \in \mathcal{F}_{\text{int}}(P^*)} \text{area}(G) d(q_T, G)}_{I \geq 0}. \quad (9)$$

The boundary term depends only on the total M :

$$B = \frac{h_T}{12} \sum_j m_{\sigma^*(j)} = \frac{h_T M}{12} = \frac{h_T}{12} \cdot \frac{4A_T}{5} = \frac{A_T h_T}{15} = \frac{V_T}{5} = V_{\text{piece}}.$$

Hence $I = 0$: every interior facet G of P^* satisfies $d(q_T, \text{aff}(G)) = 0$, i.e. $q_T \in \text{aff}(G)$.

Non-degenerate case. If $q_T \in \text{int}(P^*)$ then $d(q_T, \text{aff}(G)) > 0$ for every facet G , so P^* has no interior facets: $f(P^*) = 4$. By congruence every piece has $f = 4$, contradicting face-to-face (no shared interior facet).

It remains to exclude the degenerate case $q_T \in \partial P^*$. Every point of a convex polytope belongs to the relative interior of a unique face σ ; write $q_T \in \text{relint}(\sigma)$. The four boundary facets of P^* lie on T -face planes at distance $h_T/4 > 0$ from q_T , so σ is not a boundary facet and lies in the interior skeleton. We distinguish three cases by $\dim \sigma$.

Lemma 12.19 (Degenerate face case). *σ cannot be an interior facet of P^* .*

Proof. Suppose $q_T \in \text{relint}(G)$ for an interior facet G . Then $\text{aff}(G)$ is the supporting hyperplane of P^* at q_T . For any other facet $G' \neq G$, $q_T \in \text{aff}(G')$ forces $q_T \in G'$ (since $\text{aff}(G')$ supports P^* and $q_T \in P^*$), placing $q_T \in G \cap G'$ —a proper face of G , contradicting $q_T \in \text{relint}(G)$. Thus P^* has exactly one interior facet: $f - 4 = 1$. Double-counting interior facet-instances gives $5(f - 4)$ even, so $f - 4$ is even. Contradiction. $\square$

Lemma 12.20 (Degenerate edge case). *σ cannot be an interior edge of P^* .*

Proof. Suppose $q_T \in \text{relint}(e)$ for an interior edge e of P^* . From $I = 0$, every interior facet G satisfies $q_T \in \text{aff}(G)$; since $\text{aff}(G)$ is a supporting hyperplane of P^* and $q_T \in P^*$, we obtain $q_T \in G$. In the face lattice of a convex polytope, $\text{relint}(e) \cap G \neq \emptyset$ forces $e \subseteq G$, so every interior facet contains e . An edge of a 3-polytope is contained in exactly two facets, giving exactly two interior facets; the four boundary facet-planes, at distance $h_T/4 > 0$ from q_T , confirm that neither facet containing e is boundary. Hence $f = 6$; by congruence every piece has $f = 6$.

Each interior facet is shared with a unique neighbour, so the adjacency graph is 2-regular on 5 vertices, hence the 5-cycle C_5 . The facet G shared between P^* and a neighbour P' satisfies $q_T \in e \subset G$, so $q_T \in P'$. Iterating around C_5 shows $q_T \in P_k$ for all k .

Applying the $I = 0$ argument to each piece, all $10 = 5 \times 2$ interior facets lie in planes through q_T . The spherical link at q_T is a tiling of S^2 by five congruent digons (each piece contributes two interior-facet arcs). A digon is bounded by two great-circle arcs sharing the same pair of antipodal vertices; each spherical digon is therefore determined by a unique such pair. Adjacent digons sharing an edge share that pair, and since the edge-adjacency graph of the tiling is connected, all five digons share the same antipodal pair; hence the tiling is a hosohedron. Equivalently, all interior-facet planes contain a single line L through q_T .

Let $E = L \cap T$ and let H be the plane through the midpoint m of E perpendicular to L . Set $K = T \cap H$ (a convex polygon with at most four sides). Since L is contained in every interior-facet plane, the segment E lies in every piece and is the intersection of the two interior facets of each P_k . We claim that any isometry $\phi_k: P^* \rightarrow P_k$ maps boundary facets to boundary facets (and hence interior facets to interior facets). The four boundary facets of P^* lie on the four face-planes of T , whose six pairwise dihedral angles all equal $\arccos(1/3)$. An isometry preserves dihedral angles, so the images form four facets of P_k whose supporting planes also have all pairwise angles $\arccos(1/3)$. Among the six facet-planes of P_k , the two interior planes meet at $360^\circ/5 = 72^\circ \neq \arccos(1/3)$, so both interior planes cannot appear in such a quadruple. If exactly one interior plane were included, it would make angle $\arccos(1/3)$ with the remaining three face-planes of T ; but the four face-normals of a regular tetrahedron satisfy $n_i \cdot n_j = -\frac{1}{3}$ for $i \neq j$, so a unit vector u with $|u \cdot n_i| = \frac{1}{3}$ for $i = 1, 2, 3$ satisfies $u = \pm n_4$, forcing the interior plane to coincide with a face-plane of T —impossible since interior-facet planes pass through $q_T \in \text{int}(T)$. Thus ϕ_k maps boundary facets to boundary facets.

Consequently ϕ_k sends the two interior facets of P^* to those of P_k and their common edge E to the corresponding edge of P_k , which is again E . In particular ϕ_k fixes the midpoint m and preserves H , so the five cross-sections $S_k = P_k \cap H$ are congruent convex regions, each a sector with apex m and angular width 72° .

By congruence the five radial edges from m to the division points $d_1, \dots, d_5 \in \partial K$ all have a common length r , so the d_k lie on a circle C of radius r centred at m . The points $d_1, \dots, d_5$ split ∂K into five arcs with pairwise disjoint interiors. Each vertex of K lies in the interior of at most one arc, and K has at most four vertices, so at least one arc contains no vertex of K ; that arc is a single straight segment. Since all five outer arcs are congruent (as images of one another under the ϕ_k), every arc is a straight segment lying on one edge of K . With five straight arcs and at most four edges, two consecutive arcs share the same edge, placing three of the d_k on one line. But a line meets a circle in at most two points. Contradiction. $\square$

Lemma 12.21 (Degenerate vertex case). *σ cannot be a vertex of P^* .*

Proof. If q_T is a vertex of P^* , at least three facets meet there. Since the four boundary facet-planes are at distance $h_T/4 > 0$, all facets at q_T are interior. Each interior facet is shared with exactly one of the other four pieces, and two convex pieces in a face-to-face dissection share at most one facet (their intersection is convex, hence a single face), so P^* has at most four interior facets. Combined with at least three and the parity constraint $f - 4$ even, exactly four interior facets remain, giving $f = 8$. By congruence every piece has $f = 8$.

Each piece has four interior facets, giving a 4-regular adjacency graph on five vertices, i.e. K_5 . Every pair shares an interior facet through q_T , so all five pieces meet at q_T . Reapplying the $I = 0$ argument to each piece, every interior facet of every piece passes through q_T .

Intersect with a small sphere S_ε centred at q_T . The five links $\Lambda_k = P_k \cap S_\varepsilon$ are congruent spherical quadrilaterals (4 edges each) tiling S^2 edge-to-edge. By Euler's formula $F_s = 5$, $E_s = \frac{1}{2} \cdot 4 \cdot 5 = 10$, $V_s = 2 + E_s - F_s = 7$. In a face-to-face dissection every interior edge of the cell complex is shared by at least three cells (distinct neighbours in K_5), so every vertex of this spherical tiling has degree ≥ 3 , giving $2E_s \geq 3V_s$. But $20 < 21$. Contradiction. $\square$

Lemmas 12.19 to 12.21 exhaust every degenerate position of q_T on ∂P^* . Together with the non-degenerate case, the $(5, 5, 5, 5)$ residual is UNSAT.

Step 3: the $t \in \{2, 3\}$ residuals

Rescale areas to integer units, $A_T = 60$, $E = 48$ (natural units of $\sqrt{3}/30$). By piece congruence every piece carries the *same* canonical footprint multiset $(m_1, \dots, m_t)$ with $\sum m_i = E = 48$; for each T -face j with $k_j > 0$ there must exist a k_j -multisubset of the multiset summing to $A_T = 60$.

$t = 2$ residuals. Both $(4, 2, 2, 2)$ and $(3, 3, 2, 2)$ contain a face with $k_j = 2$; the pair-sum requirement $m_a + m_b = 60$ combined with $m_1 + m_2 = 48$ forces $(m_1, m_2) = (18, 30)$ uniquely (taking $2 \cdot 30 = 60$). Then:

- for $(4, 2, 2, 2)$ the $k_0 = 4$ face needs a 4-multisubset of $\{18, 30\}$ summing to 60, but the achievable sums are $\{72, 84, 96, 108, 120\}$, missing 60;
- for $(3, 3, 2, 2)$ the $k_0 = 3$ face needs a 3-multisubset summing to 60, but the achievable sums are $\{54, 66, 78, 90\}$, missing 60.

Both residuals are UNSAT at the canonical-multiset level.

$t = 3$ residuals. Exhaustive enumeration over ordered triples (m_1, m_2, m_3) with $\sum m_i = 48$ and $m_i \geq 1$, combined with the per- T -face contribution-sum test and a brute-force search for per-piece assignments, gives:

Residual	Multisets passing contribution test	Consistent assignments	Status
(5, 5, 3, 2)	0	–	UNSAT at canonical-multiset level
(5, 4, 4, 2)	1 (namely (6, 12, 30))	24	combinatorially SAT
(5, 4, 3, 3)	5	20	combinatorially SAT
(4, 4, 4, 3)	8	54	combinatorially SAT

Only (5, 5, 3, 2) closes at the canonical-multiset level. Pooling the contribution-sum-feasible area multisets of the other three residuals yields the finite set

$$\mathcal{M} = \{ (6, 12, 30), (6, 15, 27), (6, 18, 24), (12, 12, 24), (12, 16, 20) \} \quad (10)$$

of five integer triples of distinct T -face footprints (units of $\sqrt{3}/30$). We close each of them by a topological reduction.

Lemma 12.22 (Topological parity for $t = 3$ pieces). *Let $P_1, \dots, P_5$ be a face-to-face congruent dissection of T with $t(P) = 3$ for every piece. Then each piece has $f(P) \in \{5, 7\}$ facets.*

Proof. Let $f = f(P)$. We first count boundary facets. A convex polytope has *at most* one facet per supporting hyperplane (two coplanar facets would either coincide, contradicting the definition of a facet, or leave a lower-dimensional gap between them, contradicting convexity). Hence P has at most one facet on each of the $t = 3$ touched T -faces. On the other hand, $t(P) = 3$ means P has positive 2-area on three T -faces, so it contributes at least one facet to each of them. Thus P has exactly 3 boundary facets and $f - 3$ interior facets.

Next we count interior incidences. Two distinct convex pieces P_i, P_j can share at most one facet: in a face-to-face dissection any two pieces have convex intersection equal to a (possibly empty) face of each, and a single convex face cannot equal two different facets of either piece. Consequently each interior facet of P_i is shared with a *unique* other piece, and is counted twice in the total piece–facet incidence count on interior facets. The total is $5(f - 3)$, hence even. Since 5 is odd, $f - 3$ is even, i.e. f is odd.

Finally, convexity forces $f \geq 4$, and the one-facet-per-neighbour bound from the previous paragraph gives $f - 3 \leq n - 1 = 4$, so $f \in \{5, 7\}$. $\square$

Remark 12.23 (Tetrahedral sanity check). The excluded $f = 4$ case can be ruled out a second time by elementary arithmetic, which we record as an independent consistency check. A tetrahedral piece P with $t = 3$ must have all three boundary facets meeting at a single vertex v_P of T (the three T -face planes meet only at the four vertices of T); the three edges of P issuing from v_P therefore lie along the three edges of T at v_P , with pairwise angles 60° . Writing a, b, c for the three edge lengths, the boundary triangles at v_P have areas

$$m_1 = \frac{bc\sqrt{3}}{4}, \quad m_2 = \frac{ac\sqrt{3}}{4}, \quad m_3 = \frac{ab\sqrt{3}}{4},$$

and the volume of the resulting tetrahedron evaluates (Gram determinant) to $V(P) = abc/(6\sqrt{2})$. Eliminating a, b, c yields

$$m_1 m_2 m_3 = \frac{3\sqrt{3}}{64} (abc)^2 = \frac{27\sqrt{3}}{8} V(P)^2 = \frac{27\sqrt{3}}{8} \cdot \frac{V_T^2}{25} = \frac{A_T^3}{25} = 8640 \quad (11)$$

in the integer units of (10) (with $A_T = 60$, using $V_T = (1/3)A_T h_T$ and $h_T^2 = 2A_T/3$). On the other hand, by AM–GM any triple (m_1, m_2, m_3) with $m_1 + m_2 + m_3 = 48$ satisfies $m_1 m_2 m_3 \leq (48/3)^3 = 4096 < 8640$. The tetrahedral case is therefore arithmetically infeasible, confirming the topological exclusion of Lemma 12.22.

Lemma 12.24 (Weak Congruence Transport). *Let $P_1 = \gamma_1(P), \dots, P_5 = \gamma_5(P)$ be congruent copies of a reference polytope P under rigid motions $\gamma_i \in \text{Isom}(\mathbb{R}^3)$, and suppose pieces P_i and P_j share a facet S_{ij} which is the image $\gamma_i(F_a)$ of abstract facet F_a of P and also the image $\gamma_j(F_b)$ of abstract facet F_b of P . Then the composition $\tau_{ij} := \gamma_j^{-1}\gamma_i \in \text{Isom}(\mathbb{R}^3)$ restricts to an isometry $\tau_{ij}|_{F_a}: F_a \rightarrow F_b$ of polygons. In particular F_a and F_b share*

(i) *combinatorial type: same vertex count, same edge count;*

(ii) *metric invariants: same multiset of edge lengths, same multiset of interior angles, and the same 2-area.*

Proof. $\gamma_i(F_a) = S_{ij} = \gamma_j(F_b)$ yields $\tau_{ij}(F_a) = \gamma_j^{-1}(S_{ij}) = F_b$. Restricted to F_a , the rigid motion τ_{ij} of $\mathbb{R}^3$ is an isometry of polygons $F_a \rightarrow F_b$; combinatorial data and metric invariants of F_a are thereby transferred to F_b . $\square$

Remark 12.25 (The strong claim $\tau_{ij} \in \text{Sym}(P)$ does not hold in general). It is *incorrect* to conclude that τ_{ij} belongs to $\text{Sym}(P)$: while $\tau_{ij}(F_a) = F_b$ as polygons, one has $\tau_{ij}(P) = \gamma_j^{-1}(P_i)$, i.e. the image of P lies on the side of F_b opposite P , so $\tau_{ij}(P) = P$ would force P to be reflection-symmetric across the plane of F_b , a rare accident. (Equivalently, $\tau_{ij} \in \text{Sym}(P)$ would require $P_i = P_j$ as subsets of $\mathbb{R}^3$, impossible for distinct pieces.) The correct statement is [Lemma 12.24](#) above, which we refer to as the *weak* Congruence Transport Lemma; it already suffices to close the $f = 5$ case. The $f = 7$ argument below uses the following orbit-level corollary.

Lemma 12.26 (Orbit transport). *Keep the hypotheses of [Lemma 12.24](#) and assume moreover the following orbit-detection property of the reference tile P :*

(H-orb) *Any two facets of P with equal edge-length and interior-angle multisets lie in a common $\text{Sym}(P)$ -orbit.*

Then F_a and F_b lie in a common $\text{Sym}(P)$ -orbit on the facet set of P , and there exists $g_{ij} \in \text{Sym}(P)$ with $g_{ij}(F_a) = F_b$.

Proof. By [Lemma 12.24](#), F_a and F_b have equal edge-length and interior-angle multisets; (H-orb) gives the orbit claim, and the transitive action on the orbit supplies g_{ij} . $\square$

Remark 12.27 ((H-orb) reduces to a finite table check in our setting). In general (H-orb) is a *genericity* assumption: two facets with matching polygon data either already lie in a common $\text{Sym}(P)$ -orbit or are an “accidental” pair with larger intrinsic symmetry than the rest of P can support. In the $f = 7$ analysis below we apply [Lemma 12.26](#) to pieces P whose symmetry group is a priori bounded ($\text{Sym}(P) = 1$ or $|\text{Sym}(P)| = 2$, according to the boundary-area multiset; cf. the *Closing-the- $f = 7$ -sub-case* paragraph). With $|\text{Sym}(P)| \leq 2$ and at most four interior facets, the facet-pair / orbit-pair table has at most $\binom{4}{2} \cdot 3 = 18$ entries, and (H-orb) becomes an explicit finite check: for each pair of interior facets (F, F') with equal edge-length and interior-angle multisets, verify that either $F = F'$ or there exists $g \in \text{Sym}(P)$ with $g(F) = F'$. We discharge this check multiset-by-multiset as part of the certificate `phase26c_t3_f7_exclusion.py` (not as a genericity assumption).

Lemma 12.28 (K_5 triangle cycle-consistency). *Let $P_1 = \gamma_1(P), \dots, P_5 = \gamma_5(P)$ be a face-to-face congruent dissection with $f(P) = 7$, and assume the reference tile P has $\text{Sym}(P) = \langle \sigma \rangle \cong C_2$ whose induced action on the four interior facets of P has orbit structure $(2, 2)$. Assume also hypotheses (H-orb) of [Lemma 12.26](#) and the following triangle cocycle:*

(H-coc) *For every triple $\{i, j, k\} \subset \{1, \dots, 5\}$, the chosen orbit-transport elements $g_{ij}, g_{jk}, g_{ki} \in \text{Sym}(P)$ from [Lemma 12.26](#) satisfy $g_{ij} \cdot g_{jk} \cdot g_{ki} = \text{id}$ in $\text{Sym}(P)$.*

Then no such dissection exists.

Remark 12.29 (Heuristic reduction of (H-coc) to a topological hypothesis; conditional sketch). This remark is *not* used in the formal proof of [Conditional Theorem 12.16](#): (H-coc) is stated there as an explicit standing hypothesis. Its purpose is only to record, informally, why (H-coc) is expected to follow from the more tangible topological statement

(H-trip) For every triple $\{i, j, k\} \subset \{1, \dots, 5\}$ the triple intersection $P_i \cap P_j \cap P_k$ is non-empty.

Under (H-trip) + face-to-face-ness + convexity, a developing-map / polyhedral holonomy argument in $\mathbb{R}^3$ strongly suggests the cocycle relation of (H-coc); we sketch the argument below and *explicitly flag the analytic gap* that prevents it from being a proof.

Step 1: triangle-local reduction. Fix a triple $\{i, j, k\} \subset \{1, \dots, 5\}$. Under (H-trip) the intersection $Q_{ijk} := P_i \cap P_j \cap P_k$ is a non-empty compact convex polytope. Since the P_ℓ are distinct pieces of a face-to-face convex dissection, $\dim Q_{ijk} \in \{0, 1\}$: in a face-to-face dissection each interior 2-facet is shared by exactly two pieces, so three distinct pieces cannot all share a common 2-dimensional face (ruling out $\dim Q_{ijk} = 2$); $\dim Q_{ijk} = 3$ is ruled out by disjoint interiors of distinct pieces.

Step 2: local link around Q_{ijk} . Pick a relative-interior point $q \in Q_{ijk}$. If $q \in \text{int}(T)$, choose a closed ball $B \subset \text{int}(T)$ of small radius $\varepsilon > 0$ centred at q , small enough that B meets only the three pieces P_i, P_j, P_k and no other piece P_ℓ , $\ell \notin \{i, j, k\}$ (this is possible: the other pieces are closed sets not containing q , so q has a positive distance from their union). If $q \in \partial T$ (e.g. q is a T -vertex or lies on a T -edge), replace B by $B \cap T$: this is still a compact convex neighbourhood of q in T , bounded by a spherical polygon on ∂B together with a planar region on ∂T , both of which are topological disks. Write L for the “link” of Q_{ijk} in B (resp. $B \cap T$), namely $\partial B \cap (P_i \cup P_j \cup P_k)$ (resp. $\partial(B \cap T) \cap (P_i \cup P_j \cup P_k)$). In every case (interior vs. boundary, $\dim Q_{ijk} = 0$ vs. 1) the link L is either simply connected (interior $\dim = 0$: $L \simeq S^2$; boundary cases: L is a spherical disk or rectangle) or homotopy-equivalent to S^1 (interior $\dim = 1$: L is an annulus $S^1 \times [0, 1]$ with $\pi_1(L) = \mathbb{Z}$). In the annular case L is *not* simply connected; however, the generating loop bounds a disk in the ambient T (contract through a tubular neighbourhood of the triple-intersection edge), so it is null-homotopic in T even though it is essential in L itself. This ambient-contractibility is the additional input invoked in Step 3 below for the annular case.

Step 3: polyhedral developing-map heuristic. Write $g_{\ell m} \in \text{Sym}(P) = C_2$ for the orbit-transport element attached to the shared facet $S_{\ell m}$ ([Lemma 12.26](#)). The *heuristic claim* is that $g_{ij} \cdot g_{jk} \cdot g_{ki} = \text{id}$ in C_2 because (a) the three rigid motions $\gamma_j^{-1} \gamma_i, \gamma_k^{-1} \gamma_j, \gamma_i^{-1} \gamma_k \in \text{Isom}(\mathbb{R}^3)$ compose to the identity of $\text{Isom}(\mathbb{R}^3)$ (trivial telescope), and (b) in a flat ambient space, a “developing map” from L to $\text{Sym}(P)$ should factor through $\pi_1(L)$ whenever the lift is canonical. In the cases where $\pi_1(L) = 1$ this would give $g_{ij} g_{jk} g_{ki} = \text{id}$ directly; in the annular case $\pi_1(L) = \mathbb{Z}$ one additional compatibility (every loop around Q_{ijk} bounds a disk in T) is required.

The analytic gap. Step 3 is where a *genuine* proof is missing. Going from “rigid-motion product is the identity in $\text{Isom}(\mathbb{R}^3)$ ” to “ $\text{Sym}(P)$ -bit product is the identity in C_2 ” requires controlling the passage from $\text{Isom}(\mathbb{R}^3)$ (where the telescoping is trivial) through the orbit-transport lift [Lemma 12.26](#) (which is defined only modulo the stabiliser $\text{Stab}(F_{\ell m}^{(\ell)}) \subset \text{Sym}(P)$) into the group C_2 . In general this lift is *not* a homomorphism: the orbit-transport element $g_{\ell m}$ depends on the shared facet $F_{\ell m}^{(\ell)}$ and on (H-orb), and its C_2 -bit is determined by orientation/handedness data of P that need not factor canonically through loops in L . A complete proof would require either

- (i) citing a polyhedral developing-map lemma whose hypotheses can be verified in our setting (e.g. Thurston, *Three-Dimensional Geometry and Topology*, Ch. 3; Ratcliffe, *Foundations of Hyperbolic Manifolds*, Ch. 8), or
- (ii) proving directly that the bit map b : triangles of $K_5 \rightarrow \mathbb{Z}/2\mathbb{Z}$ coincides with the Čech coboundary of a “handedness” labelling $s: \{1, \dots, 5\} \rightarrow \mathbb{Z}/2\mathbb{Z}$, which in the C_2 case would automatically give $\delta b = 0$.

We have not carried out either reduction to our own satisfaction. Pending such a proof we *do not claim* that (H-trip) implies (H-coc); consequently, in [Conditional Theorem 12.16](#) we use (H-coc) as an explicit standing hypothesis rather than deriving it from (H-trip).

Consequences for evidence level. If the reduction $(\text{H-trip}) \Rightarrow (\text{H-coc})$ is ever formalised, the $(t = 3, f = 7, (12, 12, 24), (2, 2)\text{-orbit})$ sub-case of [Conditional Theorem 12.16](#) would move from a closure modulo (H-coc) to a closure modulo the strictly weaker (and concretely verifiable) topological statement (H-trip). We regard the formalisation of this reduction as the most immediate open problem left by the present paper.

Remark 12.30 (Status of the standing hypotheses and the logical chain). The logical chain used to close the $(t = 3, f = 7, (12, 12, 24), (2, 2)\text{-orbit})$ sub-case is

$$(\text{H-orb}) + (\text{H-coc}) \xrightarrow{\text{Lemma 12.28}} \text{no such dissection.}$$

The topological hypothesis (H-trip) is *not* part of this chain in the present paper: it is retained only as motivation for (H-coc), with an informal sketch of the expected implication recorded in [Remark 12.29](#).

All four hypotheses (H-trip), (H-coc), (H-orb), (H-Qb) are *finite* statements: each can be checked explicitly on any candidate dissection or admissible reference tile P ; none is a genericity assumption. Pending a formal proof of $(\text{H-trip}) \Rightarrow (\text{H-coc})$, the $(t = 3, f = 7, (12, 12, 24), (2, 2)\text{-orbit})$ sub-case of [Conditional Theorem 12.16](#) is at evidence level L2+. Of the seven admissible (t, k) -residuals, four are closed *unconditionally* (both $t = 2$ residuals, the $t = 3$ residual $(5, 5, 3, 2)$, and the $t = 4$ residual); the remaining three $t = 3$ residuals invoke (H-orb) and (H-Qb) uniformly (via parity reduction + $f = 5 / f = 7$ closures), and (H-coc) is additionally invoked only on the single $(12, 12, 24)/(2, 2)\text{-orbit}$ sub-case of the $f = 7$ branch.

Counting convention. The 4+3 split counts the seven (t, k) -orbits of the role-distribution census: four need no standing hypothesis; the other three (all with $t = 3$ and combinatorial SAT after the parity step) each invoke (H-orb) on their $f = 7$ branches and (H-Qb) on their $f = 5$ Qb sub-cases, with (H-coc) entering only on the $(12, 12, 24)/(2, 2)$ sub-case inside $f = 7$. The result file `phase26e_synthesis_theorem_26_1_results.json` uses the same split via the fields `unconditional_residuals` and `conditional_residuals`.

Proof of Lemma 12.28. By [Lemma 12.26](#) applied edgewise (using (H-orb)), each shared interior facet S_{ij} determines an element $g_{ij} \in \text{Sym}(P)$; by the C_2 assumption we have $g_{ij} \in \{\text{id}, \sigma\}$, encoded by a bit $b_{ij} := \text{bit}(g_{ij}) \in \{0, 1\}$.

Step 1: K_5 decomposes into two edge-disjoint 5-cycles. Let A be the set of K_5 -edges whose shared facet lies in the first $\text{Sym}(P)$ -orbit (size 2) of interior facets of P , and B the set of edges in the second. Each piece P_i has 4 interior facets, two per orbit; therefore i has A -degree 2 and B -degree 2. Hence A and B are two 2-regular graphs on 5 vertices, edge-disjoint, with $|A| = |B| = 5$. The unique 2-regular simple graph on 5 vertices is the 5-cycle C_5 (any 2-regular graph is a disjoint union of cycles, and 5 is prime so it must be a single cycle), so $A = B = C_5$ as graphs and $A \sqcup B = K_5$.

Step 2: bit parametrisation. At each piece P_i the two A -slots carry the two A -orbit elements $\{F_A, \sigma F_A\}$ of P *exactly once each* (distinct facets carry distinct abstract labels). Orient the A -cycle by a fixed cyclic order on $\{1, \dots, 5\}$ and, at each i , call “forward A -slot” the slot pointing along this orientation and “backward A -slot” the remaining A -slot; let $x_i \in \{0, 1\}$ record whether the forward slot at i carries F_A ($x_i = 0$) or σF_A ($x_i = 1$). Since each P_i uses each A -orbit element exactly once, the backward A -slot at i then necessarily carries the *complementary* element $\sigma^{1+x_i} F_A$. On the directed A -edge $i \rightarrow j$ the shared interior facet S_{ij} is the forward A -slot of i and the backward A -slot of j , yielding

- forward label at i : $\sigma^{x_i} F_A$;

- backward label at j (same physical facet S_{ij} , viewed from j): $\sigma^{1+x_j} F_A$, by the complementarity observation above applied at j .

The orbit-transport equation $g_{ij}(\sigma^{x_i} F_A) = \sigma^{1+x_j} F_A$ in $\text{Sym}(P) = C_2$ gives

$$g_{ij} = \sigma^{1+x_i+x_j}, \quad \text{i.e.} \quad b_{ij} \equiv 1 + x_i + x_j \pmod{2} \quad \text{on } A\text{-edges,}$$

and analogously $b_{ij} \equiv 1 + y_i + y_j \pmod{2}$ on B -edges using y_i for the forward B -slot. (The “+1” records that the forward slot at i and the backward slot at j necessarily use *different* orbit elements, since each piece uses each orbit element exactly once per orbit.)

Step 3: triangle reduction. Recall from Step 1 that A and B are each isomorphic to C_5 (the 5-cycle on 5 vertices). A 5-cycle is *triangle-free* (its girth equals 5): no three vertices of C_5 form a triangle in A alone, and likewise in B alone. Hence every triangle of $K_5 = A \sqcup B$ has at least one A -edge and at least one B -edge; the two admissible A/B -splits are therefore $(|A|, |B|) = (1, 2)$ or $(2, 1)$. In either case exactly two of its three vertices are incident to *both* an A -edge and a B -edge within the triangle – call them the *mixed* vertices $\{i', j'\}$. Summing $b_{ij} + b_{jk} + b_{ki} \pmod{2}$ collapses the three “+1” terms to 1 and the six “ x_r and y_r ” terms telescope, each vertex $r \in \{i, j, k\}$ contributing $x_r + y_r = z_r$ iff r is mixed and 0 otherwise. Thus

$$b_{ij} + b_{jk} + b_{ki} \equiv 1 + z_{i'} + z_{j'} \pmod{2}, \quad z_r := x_r + y_r \pmod{2}.$$

Step 4: cocycle $\Rightarrow$ pigeon-hole. (H-coc) forces the left-hand side to vanish for every triangle, i.e. $z_{i'} + z_{j'} \equiv 1 \pmod{2}$ for every mixed pair. An elementary check (carried out in `phase26c_t3_f7_exclusion.py`) shows that as $\{i, j, k\}$ ranges over the ten triangles of K_5 , the mixed-pair $\{i', j'\}$ ranges over every unordered pair of $\{1, \dots, 5\}$; consequently $z_r \neq z_s$ for every $r \neq s$, forcing the five bits z_r to be pairwise distinct – impossible in the two-element set $\{0, 1\}$. This contradicts the existence of the dissection.

Finally, a brute-force enumeration over all $2^{10} = 1024$ parametric assignments $((x_i)_{i=1}^5, (y_i)_{i=1}^5)$ (see `phase26c_t3_f7_exclusion.py`) confirms that no (x, y) produces a bit vector b satisfying all ten triangle cocycle equations. $\square$

Closing the $f = 5$ sub-case. A convex 3-polytope with $f = 5$ is either a quadrilateral pyramid (four triangles + one quadrilateral) or a triangular prism (two triangles + three quadrilaterals). A convex polygonal facet of a convex 3-polytope contained in T that lies on ∂T necessarily lies on a *single* T -face plane (by convex support of T). Each $t = 3$ piece must place *three* of its five facets on ∂T distributed over three distinct T -face planes (three by the hypothesis $t = 3$; distinct because each of the three touched T -faces receives positive 2-area from the piece), giving the five combinatorial sub-cases enumerated below. Three of the five are excluded unconditionally by elementary metric arguments. The remaining two sub-cases – Qb-adj and Qb-opp – are closed under the explicit geometric hypothesis (H-Qb) introduced below; absent (H-Qb), they remain open at the theorem-level bridge.

- *Pyramid with three lateral triangles on ∂T ; prism with one triangle and two quadrilaterals on ∂T* (unconditional closure). In both configurations the two interior facets have *mis-matched* edge-counts: one of them is a triangle and the other is a quadrilateral. By the weak [Lemma 12.24](#), at each shared interior facet the two incident pieces carry congruent abstract polygons, hence in particular equal edge-count. Thus the five triangle-slot interior facets (one per piece) pair among the five pieces into a perfect matching on the 5 vertex-labels $\{1, \dots, 5\}$, as do the five quadrilateral-slot interior facets; a perfect matching on an odd number of vertices is impossible, so these two sub-cases are excluded.
- *Pyramid with its quadrilateral base on ∂T* (case Qb; closed under (H-Qb)). Label the pyramid as $(b_1, b_2, b_3, b_4; a)$ with $b_1, \dots, b_4$ the four base vertices cyclically around the

quadrilateral on a T -face plane T_α and $\mathbf{a}$ the apex. Two of the four lateral triangles $F_i = \triangle b_i b_{i+1} \mathbf{a}$ ($i \in \mathbb{Z}/4\mathbb{Z}$) must additionally lie on ∂T ; the two remaining lateral triangles become the two interior facets. The chosen boundary pair is either *adjacent* (indices differ by 1) or *opposite* (indices differ by 2):

- **Qb-adj.** Say F_1 on T_β and F_2 on T_γ . The base edge $b_1 b_2$ lies in $T_\alpha \cap T_\beta = e_{\alpha\beta}$, the base edge $b_2 b_3$ in $T_\alpha \cap T_\gamma = e_{\alpha\gamma}$; hence $b_2 \in e_{\alpha\beta} \cap e_{\alpha\gamma} = \{v_0\}$, where v_0 is the T -vertex opposite T_δ . Similarly the apex satisfies $\mathbf{a} \in T_\beta \cap T_\gamma = e_{\beta\gamma}$, the T -edge through v_0 opposite T_α ; and $b_1 \in e_{\alpha\beta}$, $b_3 \in e_{\alpha\gamma}$, $b_4 \in T_\alpha$. The four-parameter family of Qb-adj pyramids is thus non-empty *at the level of single-piece placement*; however, a full 5-piece face-to-face congruent dissection requires in addition a compatible pairing of pieces along each interior facet.
- **Qb-opp.** Say F_1 on T_β and F_3 on T_γ . Then $b_1, b_2 \in e_{\alpha\beta}$, $b_3, b_4 \in e_{\alpha\gamma}$, $\mathbf{a} \in e_{\beta\gamma}$, so all five pyramid vertices lie on the three T -edges concurrent at v_0 . Again the single-piece family is non-empty; 5-piece global compatibility is the operative constraint.

In both **Qb-adj** and **Qb-opp**, the exclusion reduces to the global compatibility of 5 congruent copies of such a pyramid in a face-to-face dissection of T . We adopt the following explicit hypothesis and refer the proof (or disproof) to a separate combinatorial-geometric study:

- (H-Qb) No face-to-face congruent dissection of T into 5 convex pieces admits a piece of combinatorial type **Qb** (quadrilateral pyramid with base on ∂T and two lateral triangles on ∂T), in either the **Qb-adj** or **Qb-opp** sub-configuration.

(H-Qb) is a finite geometric check on the single-piece placement space (at most 4 scalar degrees of freedom for **Qb-adj**, at most 3 for **Qb-opp**, matching `phase26c_t3_f5_exclusion.py`) together with the 5-piece compatibility; its full discharge requires either a rational/SMT solver over the parameter space or a multi-piece rigidity argument not developed here. We record the status of (H-Qb) as **L1+** (strong partial evidence: the single-piece volume constraint $V(P) = V_T/5$ and the admissible boundary-area multisets of Eq. (10) cut each unconstrained single-piece family to an at most 1-parameter locus, on which 5-piece face-to-face compatibility becomes combinatorially constrained; a complete certification is deferred to future work).

- *Prism with its two triangular faces on ∂T* (unconditional closure). The two triangular faces of a prism are parallel. If both sit on T -faces, those two T -face planes must be parallel; but the four T -face planes are pairwise incident with dihedral $\arccos(1/3) \neq \pi$, so no two are parallel. Sub-case excluded.
- *Prism with its three quadrilateral faces on ∂T* (unconditional closure). The three lateral edges of a prism are pairwise parallel; if each lateral quadrilateral sits on a distinct T -face, each lateral edge lies in the intersection of two T -face planes, i.e. on a T -edge. The three T -edges containing lateral edges are pairwise incident (they meet at a common T -vertex, being the three edges issuing from the T -vertex opposite the fourth unused T -face); in particular no two of the three chosen T -edges are parallel, contradicting the prism's parallel-lateral-edges property.

Remark 12.31 (Status of (H-Qb) and honest bookkeeping). The **Qb-adj** and **Qb-opp** sub-cases are the *only* sub-cases of the $f = 5$ closure where the paper defers to a standing hypothesis; the other three sub-cases are closed unconditionally by the elementary arguments above. An explicit **Qb-adj** pyramid in the standard regular tetrahedron with vertices $(\pm 1, \pm 1, \pm 1)$ taking a sign pattern of even parity shows that the **Qb-adj** geometric family is *non-empty at the single-piece level*: e.g. $b_2 = v_0$, $b_1 \in e_{\alpha\beta} \setminus \{v_0\}$, $b_3 \in e_{\alpha\gamma} \setminus \{v_0\}$, b_4 generic in T_α , $\mathbf{a} \in e_{\beta\gamma} \setminus \{v_0\}$ yields a non-degenerate convex pyramid. Earlier drafts of this paper claimed a single-piece

degeneracy argument in Qb-adj/Qb-opp which was a notational artefact (collision between the apex symbol and the T -vertex symbol in the companion script `phase26c_t3_f5_exclusion.py`); that argument is retracted and replaced by the standing hypothesis (H-Qb).

The five-sub-case closure is certified uniformly in $\mathcal{M}$ by `phase26c_t3_f5_exclusion.py` (21/21 passing tests).

Closing the $f = 7$ sub-case. If $f(P) = 7$, each piece has 4 interior facets and the interior adjacency graph is K_5 . The $\text{Sym}(P)$ -orbit structure on the 4 interior facets is one of $(1, 1, 1, 1)$, $(2, 1, 1)$, $(3, 1)$, $(2, 2)$, (4) . Throughout this paragraph we invoke hypothesis (H-orb) on the reference tile P ; the finite facet-table check needed to discharge it is carried out case-by-case by the certificate `phase26c_t3_f7_exclusion.py`.

- *Singleton-orbit elimination.* Any orbit structure containing a singleton $\{F\}$ forces, via the weak Lemma 12.24, all five label- F facets to be mutually congruent polygons; by Lemma 12.26 they all lie in the same $\text{Sym}(P)$ -orbit of P , which by assumption has size 1. Each such facet is therefore shared between two specific pieces determined by the orbit slot; since each piece provides exactly one slot, the five slots must pair into a perfect matching on 5 vertices – impossible since 5 is odd. This eliminates $(1, 1, 1, 1)$, $(2, 1, 1)$, $(3, 1)$.
- *Distinct-boundary-area multisets.* For the four three-distinct-area multisets

$$(6, 12, 30), \quad (6, 15, 27), \quad (6, 18, 24), \quad (12, 16, 20),$$

in $\mathcal{M}$, any $g \in \text{Sym}(P)$ must permute the three boundary facets while preserving their (distinct) 2-areas; hence g fixes each boundary facet *as a set*. The three boundary facets meet at the single T -vertex v_P (the three T -face planes share only the four T -vertices), so g fixes v_P . Linearising at v_P (so $v_P = 0$ and $g \in \text{O}(3)$), the three boundary facet planes restrict to three pairwise transverse 2-planes Π_1, Π_2, Π_3 through 0, each of which g stabilises setwise. Decompose $g = \varepsilon \cdot h$ with $\varepsilon = \det g \in \{\pm 1\}$ and $h \in \text{SO}(3)$:

- An element $h \in \text{SO}(3)$ stabilising *each* of three pairwise distinct 2-planes Π_1, Π_2, Π_3 through 0 whose triple intersection is $\{0\}$ must be id. Indeed a non-trivial rotation $h \in \text{SO}(3) \setminus \{\text{id}\}$ has a unique 1-dimensional axis $L = \text{fix}(h)$; stabilising a 2-plane Π_i through 0 setwise forces $L \subseteq \Pi_i$ or $L \perp \Pi_i$ (these are the only rotation axes compatible with setwise stabilisation of a plane through 0). We split into cases according to the number of indices i with $L \subseteq \Pi_i$:

Case A ($L \subseteq \Pi_i$ for all three i). Then $L \subseteq \Pi_1 \cap \Pi_2 \cap \Pi_3 = \{0\}$, contradicting $\dim L = 1$.

Case B ($L \subseteq \Pi_{j_1}, \Pi_{j_2}$ and $L \perp \Pi_{i_0}$ for some labelling). Two distinct 2-planes Π_{j_1}, Π_{j_2} through 0 intersect in a 1-dimensional line $\ell := \Pi_{j_1} \cap \Pi_{j_2}$; since $L \subseteq \ell$ and both are 1-dimensional, $L = \ell$. The line ℓ is the linearisation at v_P of the T -edge $e_{j_1 j_2}$, so L is parallel to $e_{j_1 j_2}$. The condition $L \perp \Pi_{i_0}$ then says that the T -edge $e_{j_1 j_2}$ is perpendicular to the T -face T_{i_0} . In the regular tetrahedron, an edge e and a face T_i are *incident* iff $e \subset T_i$ (equivalently, T_i contains both endpoints of e); every edge is incident with exactly two of the four faces. It is an elementary computation that a T -edge makes the angle $\arccos(1/\sqrt{3}) \neq \pi/2$ with every *non-incident* T -face (i.e. any face not containing that edge). Here T_{i_0} is, by the standing choice of labelling, the unique third T -face through the vertex v_P distinct from the two T -faces T_{j_1}, T_{j_2} that contain the edge $e_{j_1 j_2}$; in particular $e_{j_1 j_2} \not\subset T_{i_0}$, so $e_{j_1 j_2}$ is non-incident with T_{i_0} in the sense just specified, and the orthogonality $L \perp \Pi_{i_0}$ would force the angle $\pi/2$, a contradiction.

Case C ($L \subseteq \Pi_{i_0}$ and $L \perp \Pi_{j_1}, \Pi_{j_2}$). Then L is perpendicular to both Π_{j_1} and Π_{j_2} ; two distinct planes through 0 cannot share a common perpendicular line (a plane's perpendicular is a unique line, and two distinct planes have distinct perpendicular lines). Contradiction.

Case D ($L \perp \Pi_i$ for all three i). Same argument: three distinct planes through 0 cannot share a common perpendicular line. Contradiction.

Hence $h = \text{id}$. The required input $\Pi_1 \cap \Pi_2 \cap \Pi_3 = \{0\}$ holds because Π_1, Π_2, Π_3 are the three T -face planes through the T -vertex v_P linearised at v_P , and the three T -face planes through a common T -vertex of a regular tetrahedron meet set-theoretically only at that vertex (since the fourth T -face is opposite v_P , the three are in general linear position).

- If $\varepsilon = -1$, then $-g = (-1) \cdot g \in \text{SO}(3)$ and still stabilises each Π_i setwise (since Π_i is a 2-plane through 0, $-\Pi_i = \Pi_i$). By the previous bullet $-g = \text{id}$, so $g = -\text{id}$. But $-\text{id} \notin \text{Sym}(P)$: the cone of P at v_P is contained in the open positive cone spanned by the three outward T -edge directions at v_P (since $P \subseteq T$ and v_P is a T -vertex), and $-\text{id}$ maps this open positive cone to the open negative cone, disjoint from P .

Both cases yield $g = \text{id}$, so $\text{Sym}(P) = 1$ and only orbit $(1, 1, 1, 1)$ is available – already excluded above.

- *The multiset* $(12, 12, 24)$. Two boundary facets share the area 12 and the third has area 24; any non-trivial $g \in \text{Sym}(P)$ must either swap the two area-12 facets or fix all three boundary facets individually. The second option is ruled out exactly as in the preceding bullet (three pairwise non-coplanar fixed boundary planes force $g = \text{id}$). Hence g swaps the two area-12 facets and fixes the area-24 facet setwise.

Such a g fixes v_P (the unique common vertex of the three T -face planes) and, after linearising at v_P , belongs to $\text{O}(3)$ with $g(\Pi_1) = \Pi_2$, $g(\Pi_2) = \Pi_1$, and $g(\Pi_3) = \Pi_3$ setwise. Let u_1, u_2, u_3 be the unit vectors along the three T -edges issuing from v_P (so $\Pi_r = \text{span}(u_a, u_b)$ for the two indices $\{a, b\} \neq \{r\}$). The two g 's satisfying the plane conditions are:

- (i) the bisector reflection ρ fixing u_1 and swapping $u_2 \leftrightarrow u_3$;
- (ii) its composition $-\rho$ with $-\text{id}$, sending $u_1 \mapsto -u_1$ and $u_r \mapsto -u_r$ for $\{r, s\} = \{2, 3\}$.

Option (ii) sends the positive T -vertex cone $\mathbb{R}_{\geq 0}u_1 + \mathbb{R}_{\geq 0}u_2 + \mathbb{R}_{\geq 0}u_3$ to its antipode and therefore cannot belong to $\text{Sym}(P)$ (since P lies in the positive cone at v_P). Hence $\text{Sym}(P) \leq \langle \rho \rangle = C_2$, with the bisector reflection ρ as the only non-trivial element. This rules out orbit structure (4) on the four interior facets.

The only remaining orbit structure is $(2, 2)$, which is excluded by [Lemma 12.28](#) under the combined hypotheses (H-orb) and (H-coc); both are discharged as finite checks by `phase26c_t3_f7_exclusion.py`.

The closure is certified by `phase26c_t3_f7_exclusion.py` (33/33 passing tests); the overall $t = 3$ synthesis is assembled by `phase26e_synthesis_theorem_26_1.py` (40/40 passing tests).

Synthesis of the $t = 3$ step. All four $t = 3$ residuals are UNSAT: $(5, 5, 3, 2)$ at the canonical-multiset level; $(5, 4, 4, 2)$, $(5, 4, 3, 3)$, $(4, 4, 4, 3)$ by the topological chain [lemma 12.22](#) $\rightarrow$ [remark 12.23](#) $\rightarrow$ [lemma 12.24](#) (weak CTL) $\rightarrow$ [lemma 12.26](#), with [lemma 12.28](#) additionally invoked only on the single $(12, 12, 24)/(2, 2)$ -orbit $f = 7$ sub-case, which appears in the residuals $(5, 4, 3, 3)$ and $(4, 4, 4, 3)$ but *not* in $(5, 4, 4, 2)$ (whose canonical multiset $(6, 12, 30)$ falls entirely in the distinct-boundary-area sub-branch of $f = 7$ and is killed using (H-orb) alone, via the singleton $\text{Sym}(P) = 1$ argument; see the $f = 7$ closure below). The auxiliary hypothesis (H-orb)

is thus used uniformly across all three parity-reduction residuals, while (H-coc) is used only on the $(12, 12, 24)/(2, 2)$ -orbit sub-case; both are verified on a case-by-case finite-table basis by `phase26c_t3_f7_exclusion.py`.

Conclusion of Step 3 and proof of [conditional theorem 12.16](#)

Steps 1, 2 and 3 close the $t = 1$ elimination and the seven admissible (t, k) residuals respectively. Hence no face-to-face congruent dissection of T into 5 convex pieces can exist, and [conditional theorem 12.16](#) is proved. $\square$

Remark 12.32 (Convexity hypothesis for $n = 5$). Convexity is essential to the canonical-multiset argument: a non-convex piece could touch a single T -face with multiple disjoint footprints, breaking the structural claim “footprint-multiset = t nonzero values” which underlies Step 3 above.

Remark 12.33 (Comparison with $n = 16$). The $n = 5$ closure is unexpectedly clean compared with the ~ 25 sub-phases of the C24-chain at $n = 16$. The structural reason is that at $n = 16$ the per- T -face filter (8) admits the all-one-face configuration ($16/4 = 4 \geq 1$) as combinatorially feasible; deep geometric arguments (apex collapse, perpendicular-bisector planes, F_4 pairing) are then needed. At $n = 5$ the *same* filter ($5/4 > 1$) kills the $t = 1$ residual outright, reducing the problem to $t \geq 2$; once there, the smallness of 5 pieces and 4 faces pins the canonical footprint multiset to a finite list of integer tuples. For $t = 2$ and for the residual $(5, 5, 3, 2)$ at $t = 3$ elementary arithmetic on the per- T -face contribution sums suffices; the remaining three $t = 3$ residuals are closed by the topological parity reduction [lemma 12.22](#) combined with the weak Congruence Transport Lemma [lemma 12.24](#), the orbit-transport corollary [lemma 12.26](#) (under hypothesis (H-orb)) and the K_5 triangle cycle-consistency obstruction [lemma 12.28](#) (under hypothesis (H-coc)); the $f = 7$ sub-cases discharge (H-orb) and (H-coc) via finite checks in `phase26c_t3_f7_exclusion.py`, while the Qb sub-cases of $f = 5$ discharge (H-Qb) via `phase26c_t3_f5_exclusion.py`. This template (role-distribution + canonical-multiset + topological closure) transfers directly to $n \in \{7, 9, 10, 11, 13, \dots\}$, the remaining small non-divisors of 24.

12.6 Summary of the programme

[conjecture 1.2](#) is equivalent to [conjecture 1.1](#) combined with [Corollary 7.4](#). Its current evidence level is L1+, supported by

- one L2 theorem ([lemma 12.2](#), Stabiliser Transport);
- one L1+ proposition ([proposition 12.6](#), Conditional Separation, conditional on Hypothesis H);
- one conjecture at L1 ([conjecture 12.4](#), Boundary-Normal Orbits);
- a Spherical Link Program constraining pieces-per-vertex to $k \in \{1, 3, 6, \dots\}$ within a uniform convex model under Hypothesis H;
- the dihedral rigidity programme ([Section 12.1](#)) reducing [conjecture 1.2](#) to GAPs A, B;
- the $n = 16$ all-one-face conditional reduction ([theorem 12.7](#) and [conditional theorem 12.8](#)), L3 in the Coxeter-pure case and L2 (conditional on $(H_{\text{roles}}) + (H_{\text{slot}})$) in the general convex case;
- and the $n = 5$ convex impossibility modulo four explicit finite standing hypotheses (H-trip), (H-coc), (H-orb), (H-Qb) ([conditional theorem 12.16](#)), the first non-divisor of 24 *conditionally reduced* – not closed – in the convex category: four of the seven (t, k) -residuals

of the role-distribution census are *unconditional* (both $t = 2$ residuals, the $t = 3$ residual $(5, 5, 3, 2)$, and the $t = 4$ residual); the remaining three $t = 3$ residuals invoke (H-orb) and (H-Qb) uniformly, with (H-coc) additionally invoked only on the single $(12, 12, 24)/(2, 2)$ -orbit $f = 7$ sub-case. Each hypothesis is a finite check on any candidate dissection (topological for (H-trip), algebraic for (H-coc), (H-orb), geometric for (H-Qb)); (H-trip) is motivational for (H-coc) and not used directly in the proof. Unconditional closure of $n = 5$ requires completing the developing-map reduction $(\text{H-trip}) \Rightarrow (\text{H-coc})$ sketched in [Remark 12.29](#), or discharging (H-orb), (H-Qb) directly; this is the natural next step of the programme.

13 Further questions

Question 13.1 (Hinged dissections). Can any of the eight known dissections be realised as a physical hinged mechanism? Edge-hinges (1-DOF) are natural candidates for S_2 and S_4 , vertex-hinges (3-DOF) for the dense configurations S_{24} . No results are known in this direction. Demaine’s universality theorems on scissors-congruent polyhedra [5] apply to pairs of polyhedra, not to partitions of a single polyhedron; their adaptation to the present setting is open.

Question 13.2 (Higher-dimensional extension). For the regular d -simplex with symmetry group S_{d+1} of order $(d+1)!$, does n admit a congruent dissection if and only if $n \mid (d+1)!$? The answer is *no* for $d = 2$: the equilateral triangle admits congruent dissection into any $n = k^2$ pieces via self-similar subdivision. The tetrahedron ($d = 3$) may be the critical dimension at which the Dehn invariant first constrains the admissible n ; this should be a theorem for $d = 3$, with modifications for $d \geq 4$.

Question 13.3 (Burnside ring extensions). The Burnside ring framework of [Section 9](#) opens questions in *equivariant K-theory* of simplicial complexes and connections to the representation ring $R(T_d)$. The relationship between the mark table and the character table of S_4 may encode deeper number-theoretic structure.

A Computational reproduction sketch

All metric, Dehn, stabiliser, MILP and SAT-style certificates cited in the paper are produced by exact rational arithmetic using Python’s built-in `Fractions` module. We include below a minimal self-contained reference implementation for the Dehn-invariant verification at $n = 24$ (orthoscheme). The full test suites for [conditional theorem 12.8](#) and [conditional theorem 12.16](#) will be made available upon submission in the data and code capsule described in [Section 1](#).

Listing 1: Minimal SymPy verification of the Dehn scaling $D(S_{24}) = D(T)/24$ for the orthoscheme piece.

```

1 import sympy as sp
2 from sympy import Rational, Point3D, sqrt
3 # Regular tetrahedron T with edge length 1
4 A = Point3D(0, 0, 0)
5 B = Point3D(1, 0, 0)
6 C = Point3D(Rational(1, 2), sqrt(3)/2, 0)
7 D = Point3D(Rational(1, 2), sqrt(3)/6, sqrt(6)/3)
8 # Canonical orthoscheme piece S_24
9 M_AB = (A + B) / 2
10 G_ABC = (A + B + C) / 3
11 G_T = (A + B + C + D) / 4
12 S24 = [A, M_AB, G_ABC, G_T]
13 # Edge enumeration, dihedral-angle computation via dot products of
14 # outward face normals, and Dehn tensor product in R (x)_Z (R/pi Q)
15 # follow directly; total runtime < 1 s.

```

B Evidence-level glossary

- L0** *Heuristic.* A claim supported by plausibility or partial checks but without a formal proof attempt.
- L1** *Partial evidence.* A claim supported by substantial evidence (e.g. computational enumeration up to a fixed bound) but lacking a closed proof.
- L1+** *Strong partial evidence.* A claim supported by a partial proof together with the explicit identification of the remaining residual gap(s); typically a reduction to a named sub-lemma.
- L2** *Conditional theorem.* A rigorously proved implication whose conclusion depends on an explicit, named auxiliary hypothesis (e.g. convexity, face-to-face, or a named hypothesis H).
- L2+** *Finite-check conditional theorem.* Like L2, but each standing auxiliary hypothesis is a finite, explicitly verifiable condition on a candidate configuration (e.g. a cocycle identity, an orbit table, or a single-piece combinatorial type exclusion).
- L3** *Unconditional theorem.* A fully proved statement with no auxiliary hypothesis beyond those of the ambient setting (convex face-to-face congruent dissection).

C The 24-cell and quaternionic action

This appendix provides a *secondary* algebraic viewpoint. The core results of the paper (Section 3–Section 9, Section 10–Section 11) do not depend on it; it serves as a partial independent verification of the dissection inventory through the quaternionic / 24-cell correspondence, and may be skipped on first reading.

C.1 The binary tetrahedral group $2T$

The binary tetrahedral group $2T \subset S^3 \subset \mathbb{H}$ consists of 24 unit quaternions: the 8 *type-1* units $\{\pm 1, \pm i, \pm j, \pm k\}$ and the 16 *type-2* units $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$. These are precisely the 24 *Hurwitz units*, the vertices of the regular 24-cell inscribed in S^3 . The group has seven conjugacy classes of sizes $(1, 6, 1, 4, 4, 4, 4)$. Its centre is $\{\pm 1\}$, and

$$1 \longrightarrow \{\pm 1\} \longrightarrow 2T \xrightarrow{\pi} A_4 \longrightarrow 1 \quad (12)$$

is exact.

Table 15: Subgroups of $2T$ and dissection matches.

$2T$ -subgroup	$ H $	$[2T : H]$	$\pi(H) \leq A_4$	$ \pi(H) $	Dissection
$\{e\}$	1	24	$\{e\}$	1	S_{24}
$\mathbb{Z}_2$ (centre)	2	12	$\{e\}$	1	S_{12}
$\mathbb{Z}_3$	3	8	$\mathbb{Z}_3$	3	S_8
$\mathbb{Z}_4$	4	6	$\mathbb{Z}_2$	2	S_6
$\mathbb{Z}_6$	6	4	$\mathbb{Z}_3$	3	S_4
Q_8	8	3	V_4	4	S_3
$2T$	24	1	A_4	12	S_1

C.2 Triple bijection

The quaternionic conjugation action $v \mapsto qv\bar{q}$ on pure quaternions v yields a canonical triple bijection

$$\underbrace{24 \text{ Hurwitz units}}_{2T} \longleftrightarrow \underbrace{24 \text{ permutations}}_{T_d \cong S_4} \longleftrightarrow \underbrace{24 \text{ orthoschemes}}_{\text{flags}(T)}. \quad (13)$$

Proper rotations give $24 \twoheadrightarrow 12$ even permutations (A_4) with kernel $\{\pm 1\}$; composition with a fixed involution σ recovers the 12 odd permutations ($S_4 \setminus A_4$); and each of the 24 orthoschemes in the barycentric subdivision of T is hit exactly once.

C.3 Subgroup lattice and the $n = 2$ gap

The subgroup lattice of $2T$ maps under π to the subgroup lattice of $A_4 \leq S_4$. Seven of the eight dissection numbers are recovered directly from coset spaces $2T/H$: $\{1, 3, 4, 6, 8, 12, 24\}$ (Table 15). The missing case $n = 2$ arises because $2T$ has no subgroup of order 12; indeed π has image $A_4 \leq S_4$ of index 2, and all $|\pi(H)|$ divide 12. The $n = 2$ dissection is captured by the S_4 -based Burnside analysis (Section 9) via stabiliser A_4 ; see Table 15.

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